SLIIT UNI

THE KNOWLEDGE UNIVERSITY

05 – Internet Protocol (IPv4)

IE2004 Computer Networks

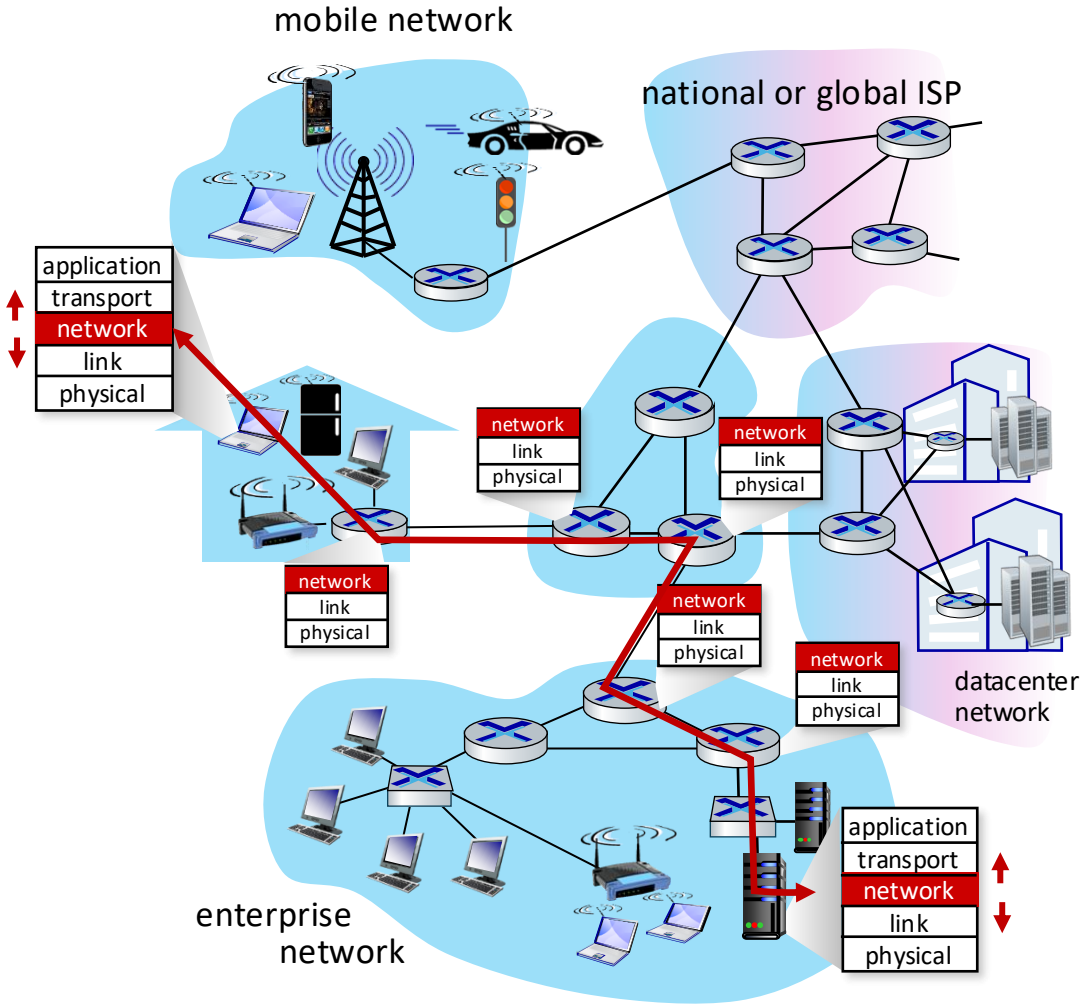
2026 July

Lesson outline

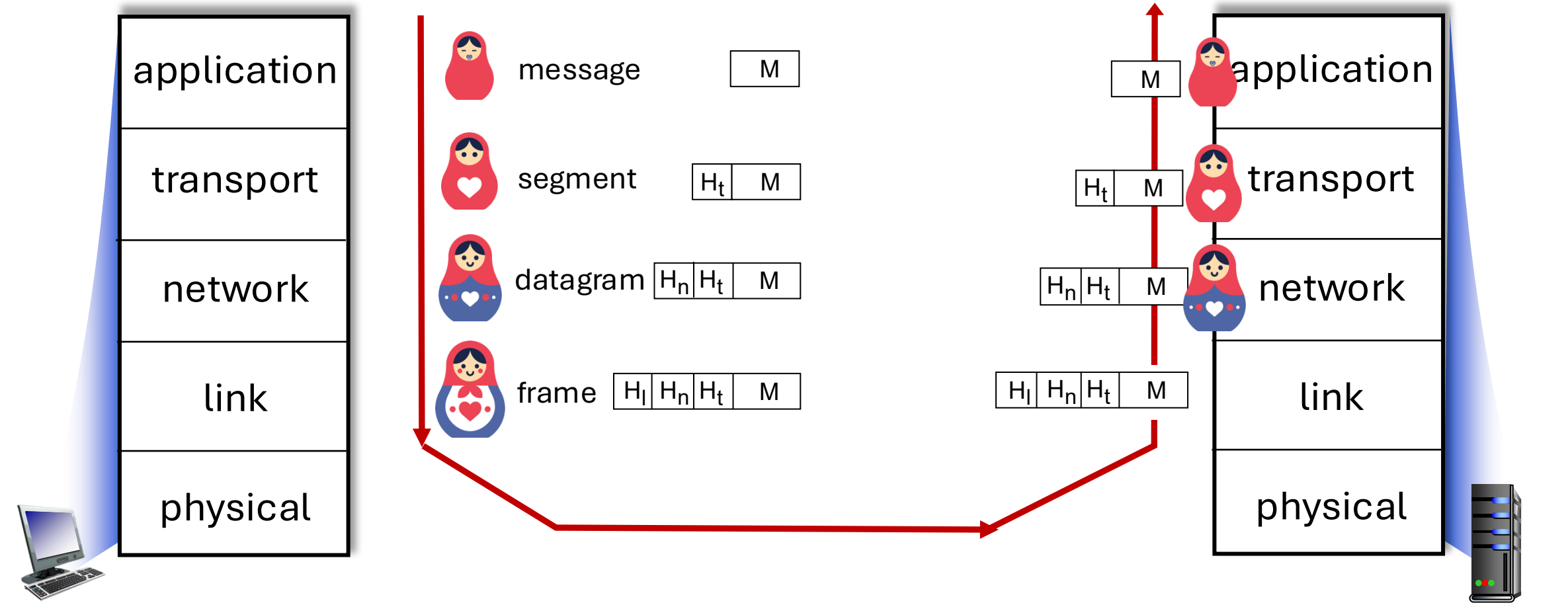
- Network layer functions
- Protocols used at the network layer
- IPv4 - IP protocol version 4: datagram format
- IPv4 fragmentation and reassembly
- IPv4 addresses
- IPv4 subnetting and address allocation
- Network Address Translation (NAT)
- Internet Control Message Protocol (ICMP)
- Address resolution Protocol (ARP)

Network layer services and protocols

- Network layer is responsible for delivering transport layer segments from sending host to receiving host
  - sender: encapsulates segments into datagrams, passes to link layer
  - receiver: delivers segments to transport layer protocol
- Network layer protocols used in *every Internet device*: hosts, routers
- Routers:
  - Examine header fields of all IP datagrams passing through
  - Moves (forwards) datagrams from one port (input) to another port (output) to deliver datagrams along the path from sender to receiver



Services, Layering and Encapsulation



This slide is from topic 2 (Protocol Architecture)

Two key network-layer functions

network-layer functions:

- *forwarding:* move packets from a router's input link to appropriate router output link
  - Done by individual routers along the path from source to destination
- *routing:* determine route taken by packets from source to destination
  - Done for by *routing algorithms*

analogy: taking a trip

- *forwarding:* process of getting through a single interchange or junction
- *routing:* process of planning a trip from source to destination

Note: Often the terms routing and forwarding are used interchangeably



forwarding



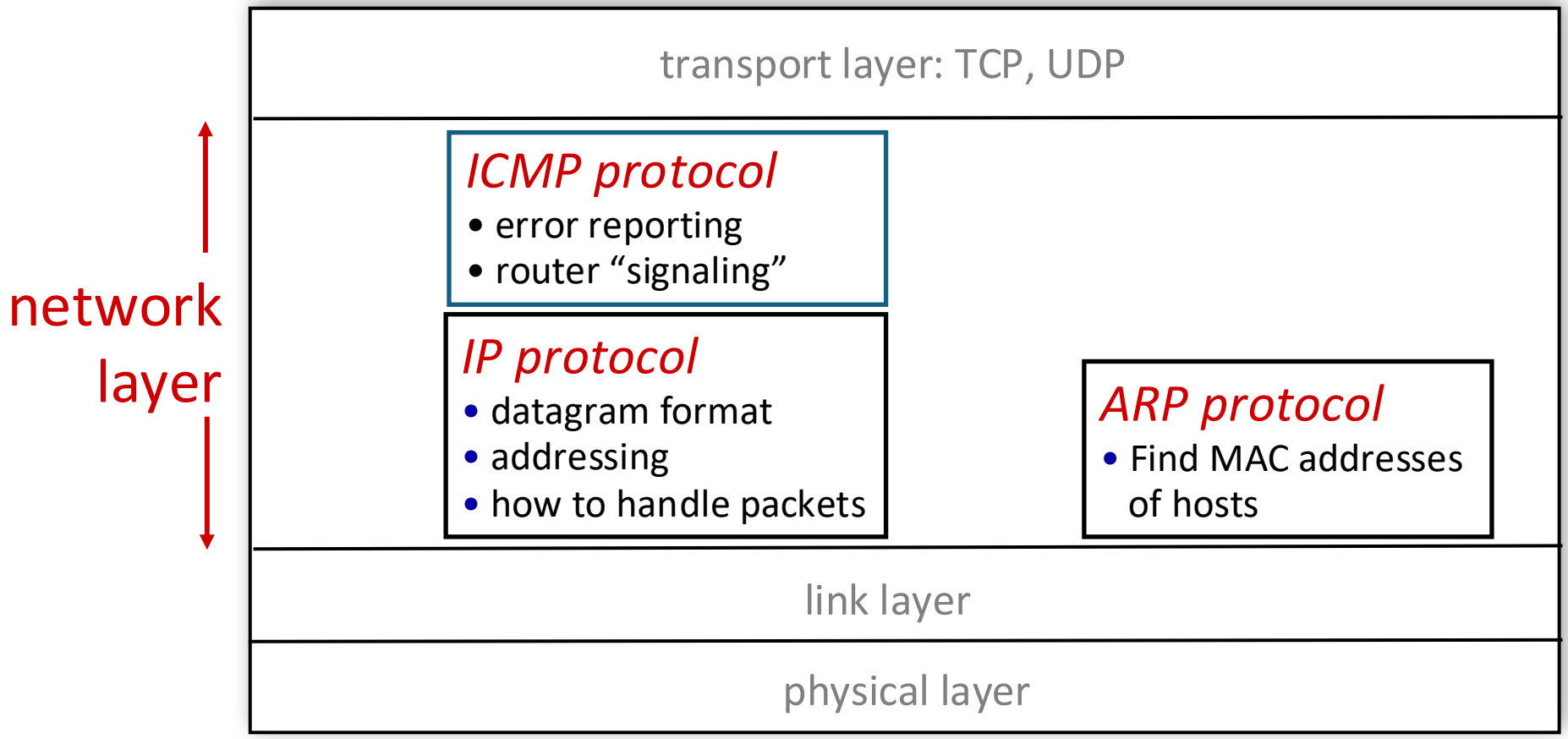
routing

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Network Layer: Internet

Protocols used in the network layer of hosts and routers:



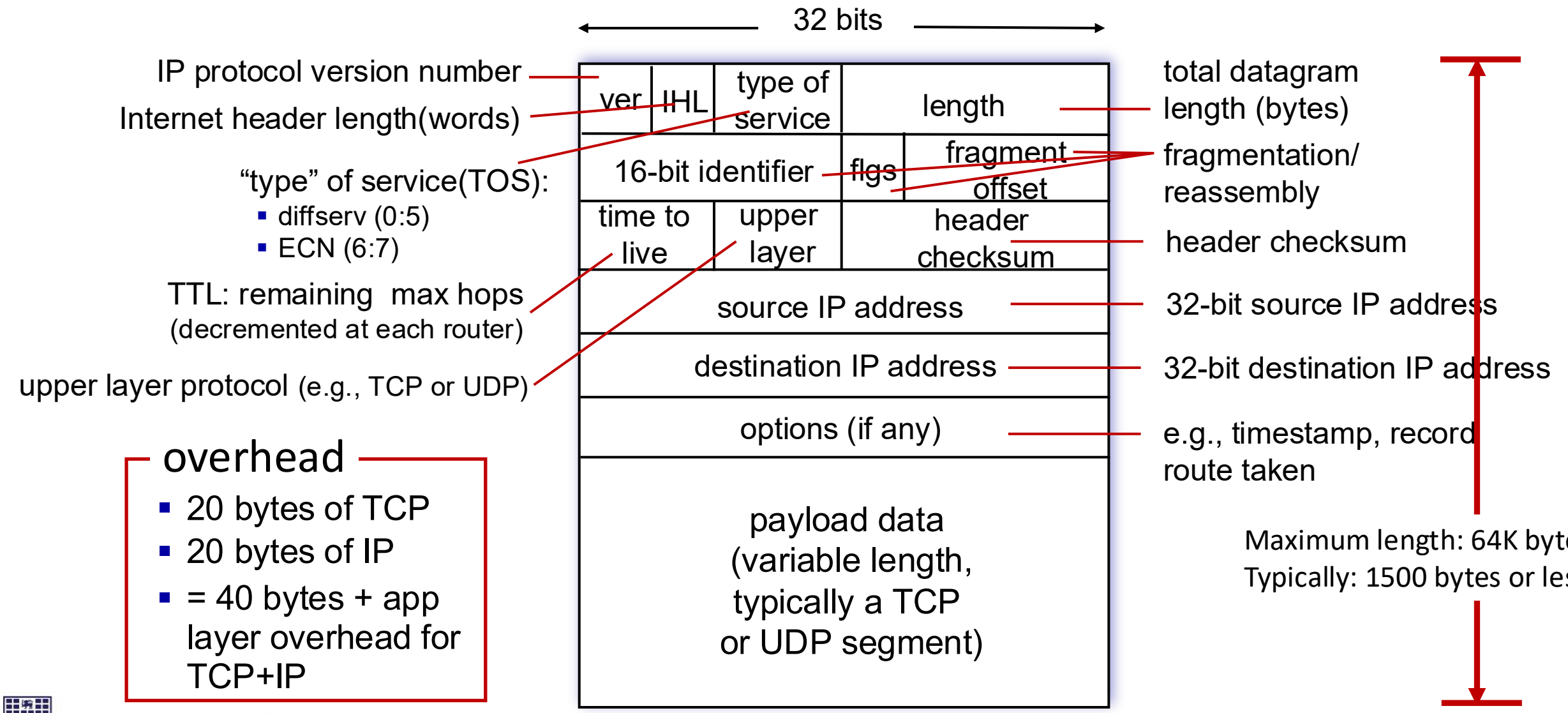
Protocols used in IP networks

- Datagram delivery system on IP networks is an unreliable, best-effort, and connectionless
  - Unreliable: Delivery is not guaranteed
  - Best effort: Some datagrams may be dropped due to congestion etc.
  - Connectionless: Each datagram is treated independently from others
- The protocol used for data transfer is the Internet Protocol (IP)
- The ICMP (Internet Control Message Protocol) is used for control
- The ARP (Address Resolution Protocol) is used for finding link layer addresses (MAC addresses) of hosts
- IP, ICMP, ARP and a few other protocols are used in the network layer

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IP Datagram format

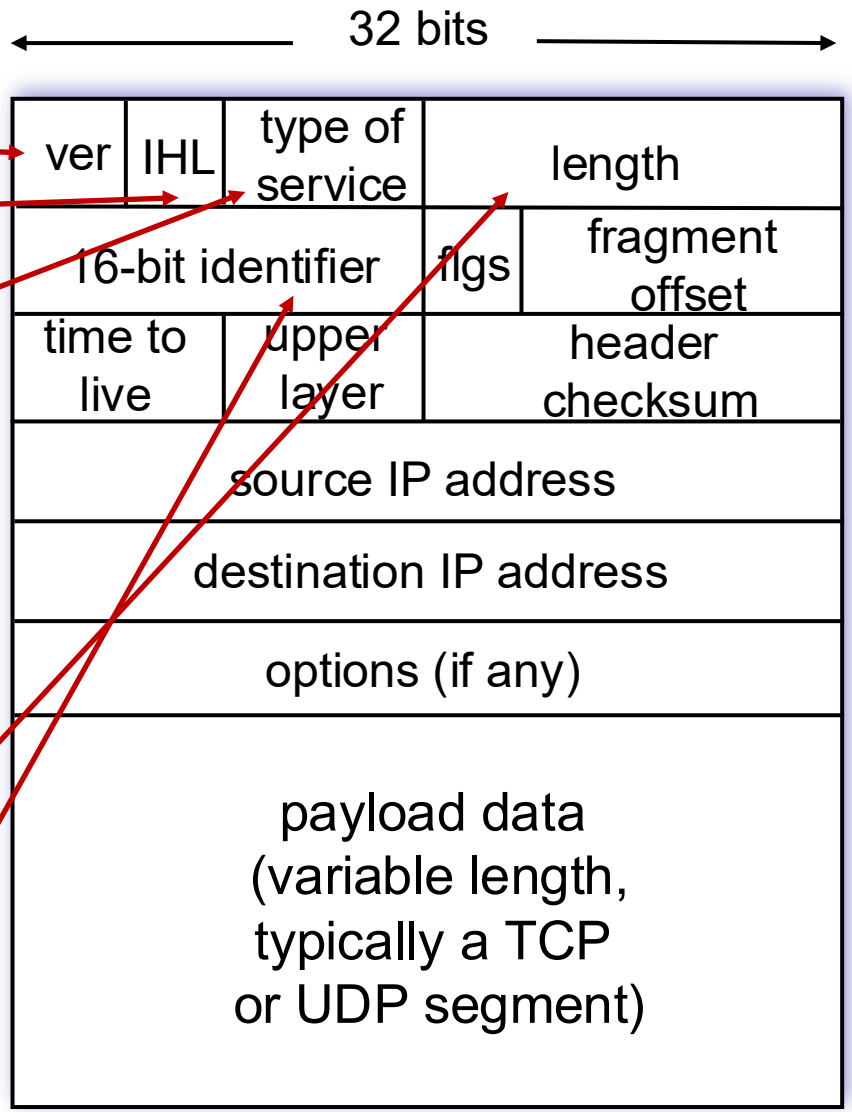


overhead

- 20 bytes of TCP
- 20 bytes of IP
- = 40 bytes + app layer overhead for TCP+IP

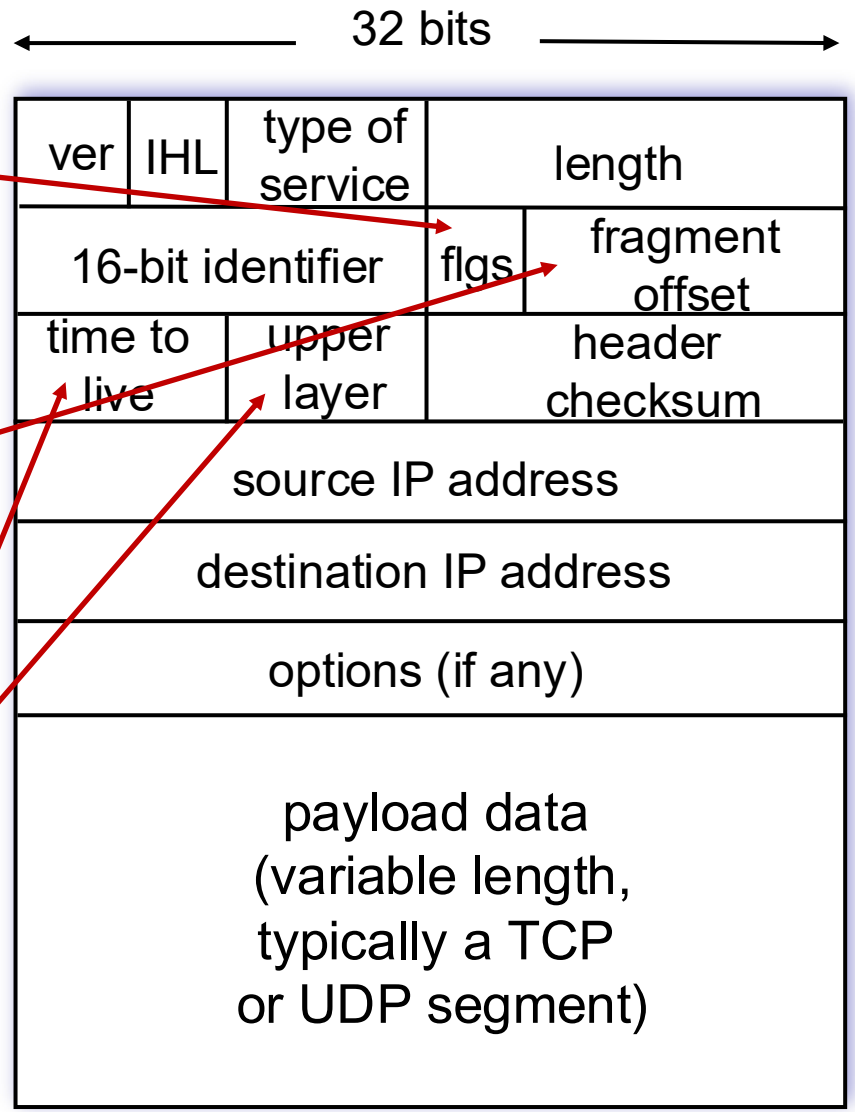
Fields of an IP datagram

- Version (4 bits): 0100 for IPv4
- Internet Header Length (IHL) (4 bits): The length of the header in 32-bit words
- Type of Service (TOS) (8 bits): The type of service desired by the sender
  - Precedence (3 bits): Indicates the priority or the importance of the datagram (7 is highest)
  - Delay (1 bit): If set to 1 low delay is desired
  - Throughput (1 bit): If set to 1 high throughput is desired
  - Reliability (1 bit): If set to 1 high reliability is desired
- Total Length (16 bits): The total length of the datagram in octets
- Identification (16 bits): A sequence number that (together with SA, DA, and Protocol fields) will uniquely identify the datagram



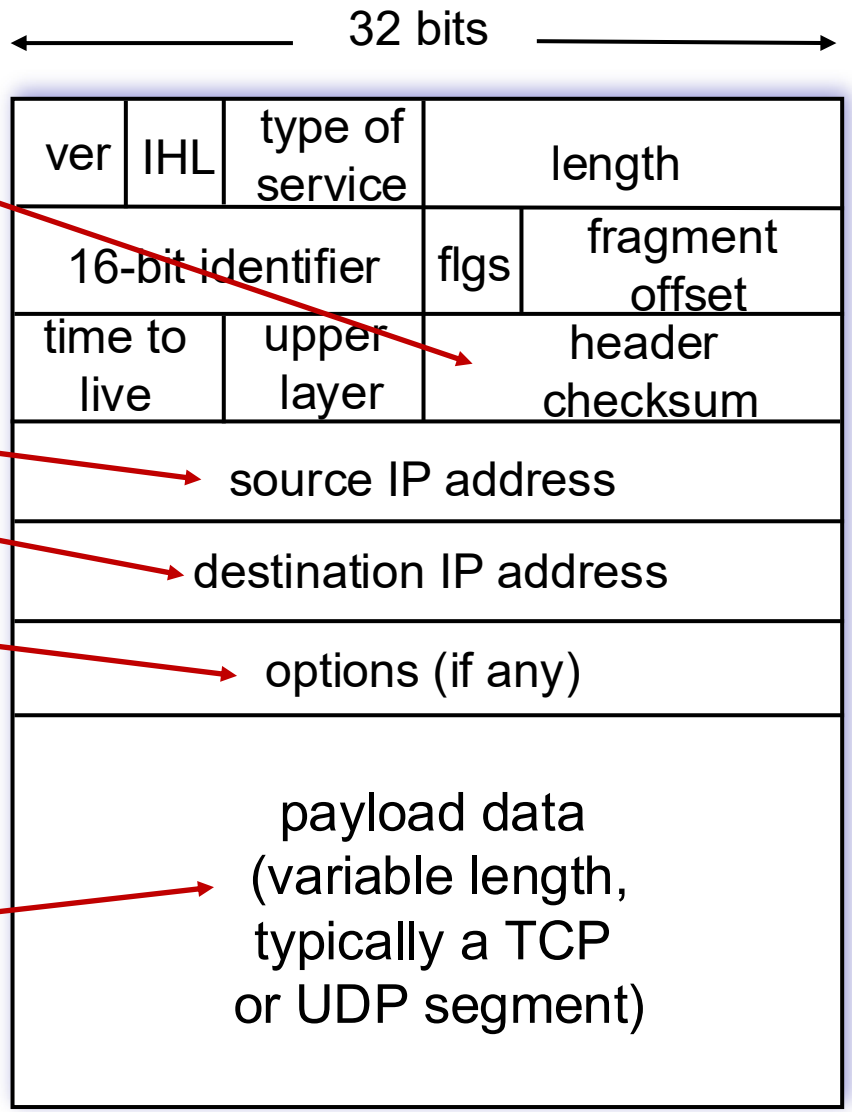
Fields of an IP datagram (2)

- Flags (3 bits): Only two flags are used at present
  - More Fragments (MF) (1 bit)
  - Don't Fragment (DF) (1 bit)
  - Unused (1 bit)
- Fragment offset (13 bits): Indicates the location of this fragment in the original datagram, measured in 64-bit units (explained later)
- Time To Live (TTL) (8 bits): Indicates the maximum number of hops that the datagram may traverse on the Internet
  - Each router should decrement the TTL by 1
- Protocol (8 bits): Specifies the higher layer protocol that is to receive this datagram at the receiver
  - 1=ICMP, 6=TCP, 17=UDP, …



Fields of an IP datagram (3)

- Header checksum (16 bits): Used for error control. Computed as follows
  - Treat the header as a sequence of 16 bit integers (assume header checksum=0 for this computation)
  - Add the integers using 1's complement arithmetic
  - Take the 1's complement of the result
- Source Address (SA) (32 bits): IP address of the sender
- Destination Address (DA) (32 bits): IP address of the recipient
- Options (variable): Allows extensions of the IP protocol (e.g. Record route, Timestamp). Length depends on the options selected
- Padding (variable): Filled with 0s to make the header length an exact multiple of 32-bit words). Not needed if no options.
- Data (variable): The actual data
  - Must be an integer multiple of octets
  - Maximum size is 65,515 octets

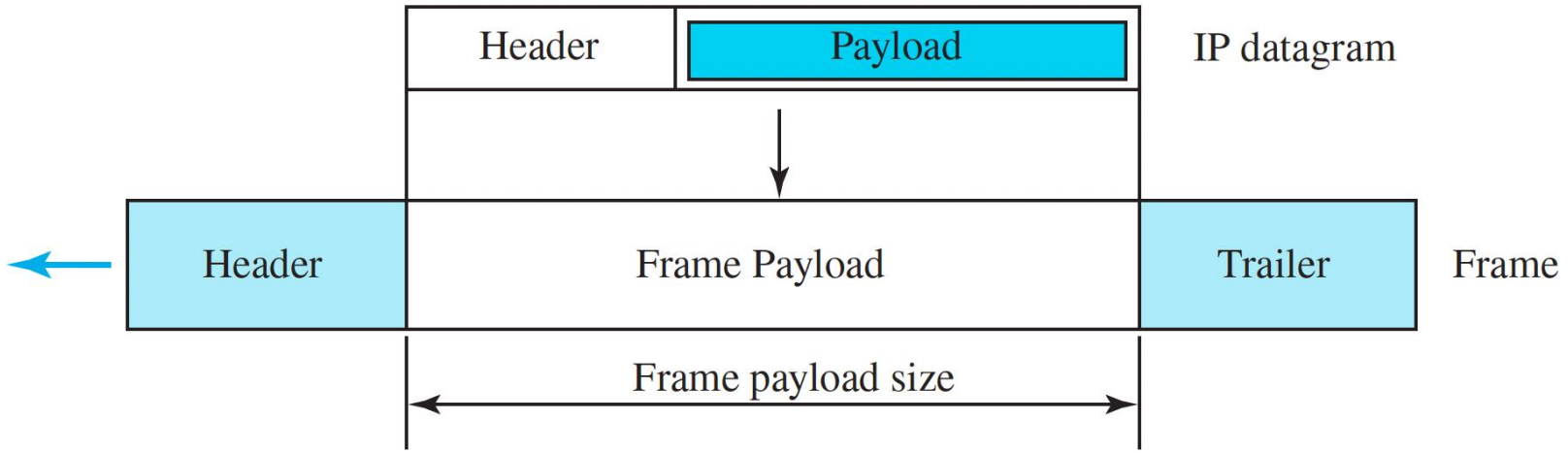


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MTU of Link layers

- Many link layer protocols limit the length of the payload section of the frame
  - This is called Maximum Transfer Unit or MTU
  - As a result, the payload of an IP datagram is limited to MTU – IP Header Length (The IP datagram standard header length is 20 bytes, with no Options)
  - For Ethernet, MTU is 1500 bytes, and payload of an IP datagram is limited to 1480 bytes
- But the length of an IP datagram can be (2^{16} -1= 65535) because the length field of the IP header is 16 bits
- How do we fit large IP datagrams in links with low MTU?
  - Fragmentation and Reassembly



MTU: Maximum size of frame payload

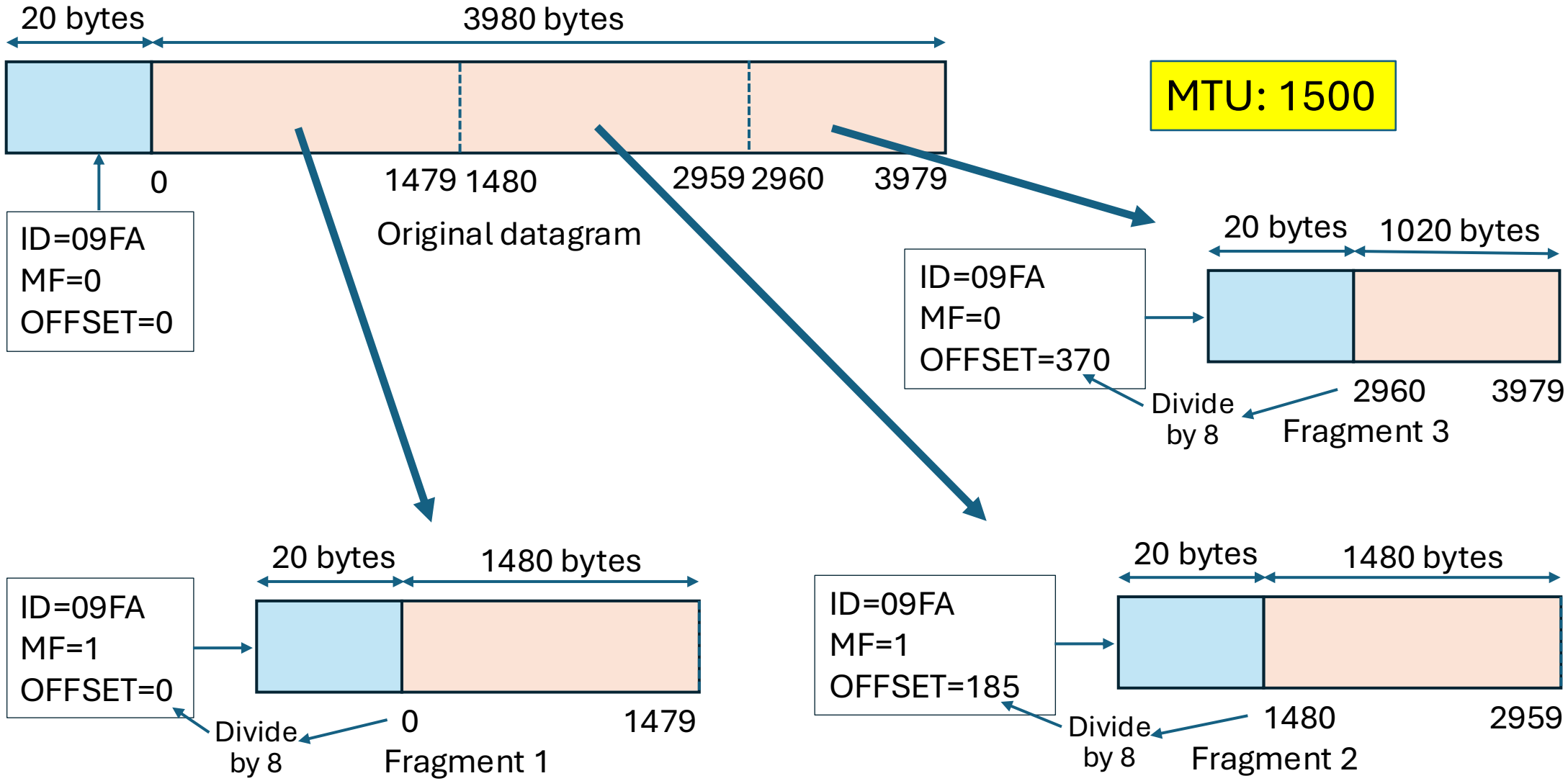
Fragmentation and reassembly

- If an IP datagram is larger than the MTU of the link on which the datagram is going out, the datagram must be fragmented
  - Fragmentation can happen at the sending host, or at any of the routers along the path to the destination host
  - A fragmented datagram may itself be fragmented if it encounters a network with an even smaller MTU
  - So, a datagram may be fragmented several times before it reaches the final destination
- The reassembly of fragments is done only at the destination host
  - each fragment becomes an independent datagram and can travel through different routes
  - So, it is only possible do the reassembly at the final destination
- IP header fields and flags used for fragmentation and reassembly
  - Identification (16 bits)
  - Fragment Offset (13 bits)
  - Don’t Fragment (DF) bit in the Flags field
  - More Fragments (MF) bit in the Flags field

Fragmentation and reassembly example

- Incoming IP datagram has ID=09FA (Hexadecimal), Length = 4000, MTU of the outgoing link = 1500
  - Payload of the incoming IP datagram = 4000 – 20 = 3980
  - Maximum possible payload size of fragments = 1500 – 20 = 1480
  - Break the incoming payload into three fragments of size 1480, 1480 and 1020 (i.e 3980 = 1480 + 1480 + 1020)
  - fragment offset is the position of the first byte in the fragment divided by 8
    - For fragment 1 it is 0 / 8 = 0
    - For fragment 2 it is 1480 / 8 = 185
    - For fragment 3 it is 2960 / 8 = 370
  - More Fragments (MF) field is set to 1 for all fragments except the last one
  - ID field of the fragments is set to ID field of the incoming datagram (= 09FA)

Fragmentation and reassembly example (ctd)



Review questions

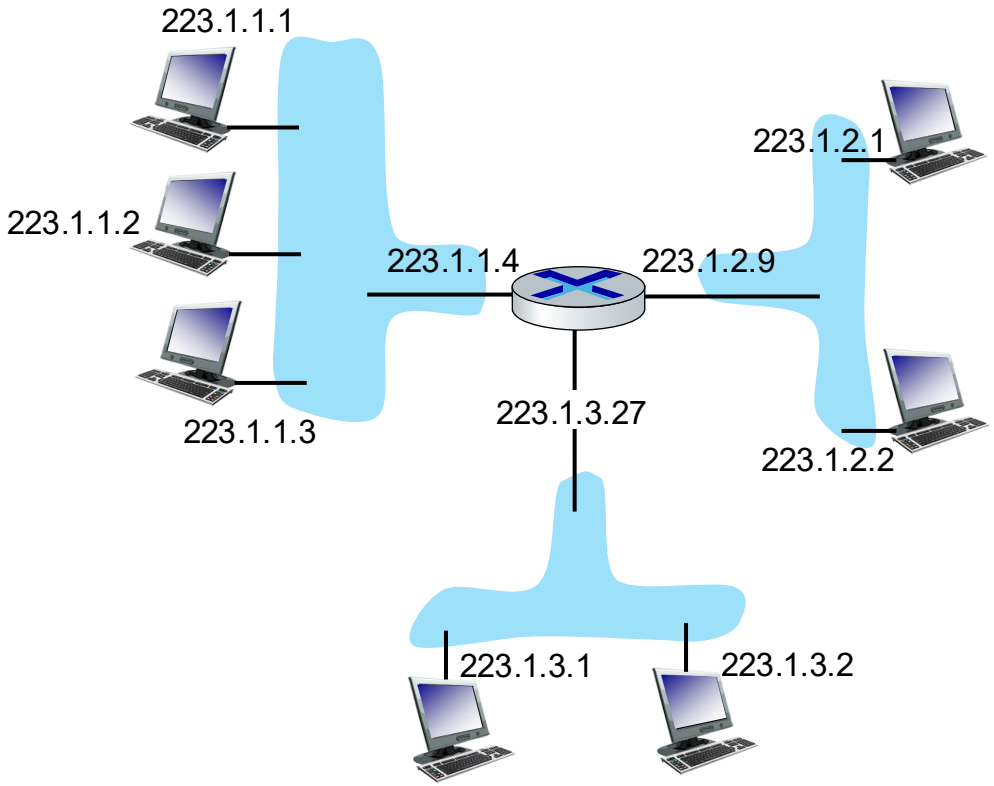
- Describe the fields in the IPv4 header and explain the function / purpose of each field
- Explain why fragmentation is needed in IPv4, and provide an example of how a large IPv4 datagram may be fragmented.

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IP addressing: introduction

- **IP address:** 32-bit identifier associated with each host or router *interface*
- **interface:** connection between host/router and physical link
  - routers typically have multiple interfaces
  - hosts typically have one or two interfaces (e.g., wired Ethernet, wireless 802.11)



dotted-decimal IP address notation:

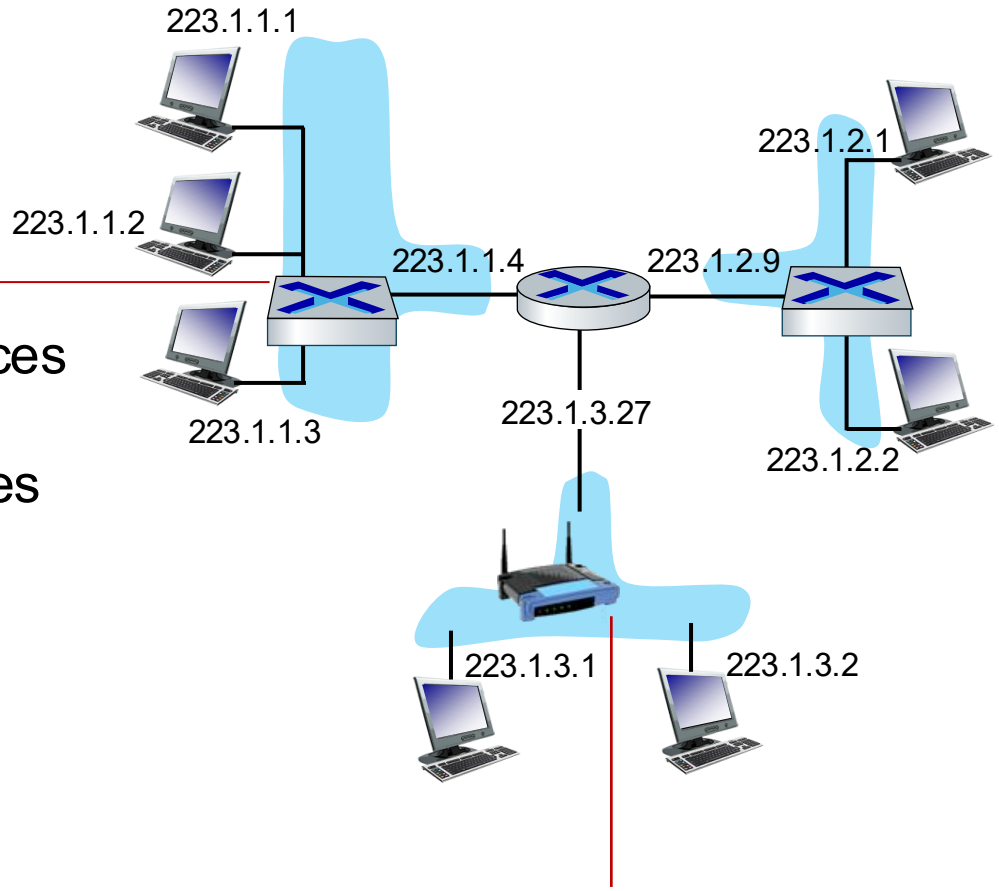
223.1.1.1 = 11011111 00000001 00000001 00000001

223 1 1 1

IP addressing: introduction

Q: how are interfaces actually connected?

A: Via a switch or a wireless access point (base station)

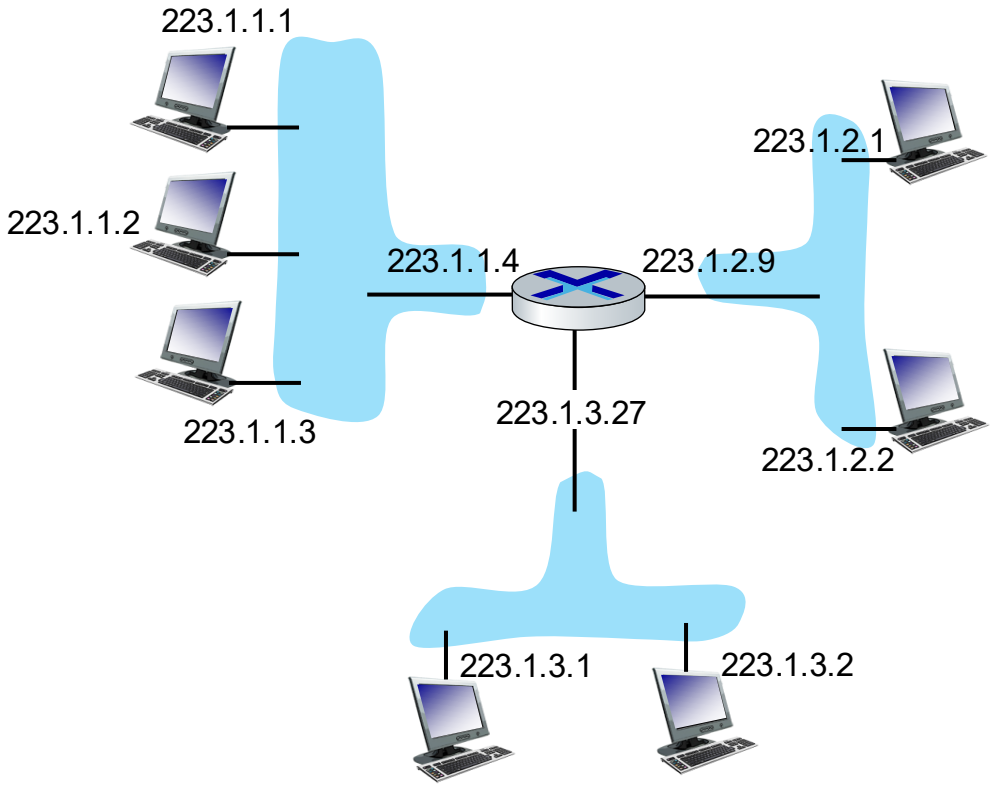


A: wired Ethernet interfaces connected by Ethernet switches

A: wireless WiFi interfaces connected by WiFi base station

Subnets

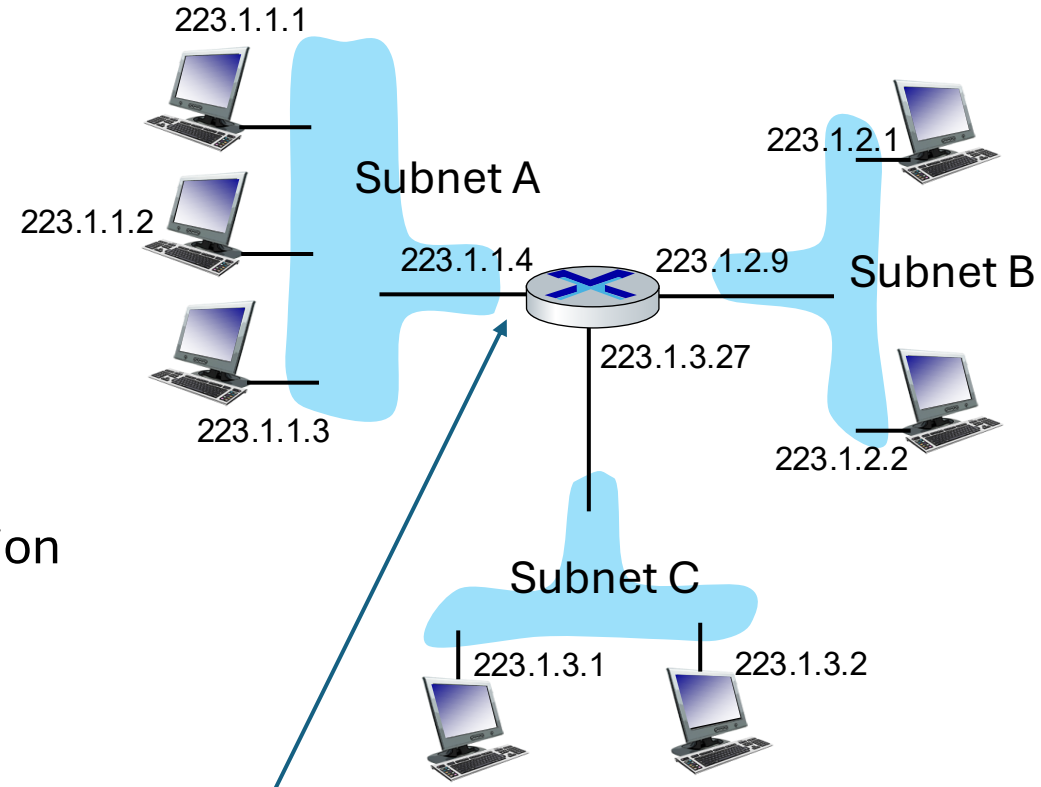
- *What's a subnet ?*
  - device interfaces that can physically reach each other without passing through an intervening router
- The term subnet is short for subnetwork, and historically meant that it is a part of a larger IP network
- But now the term is used to refer to any IP network, even if it is not a part of a larger IP network
- So, in current use: subnet = IP network



network consisting of 3 subnets

Host configuration

- A host must have an IP address to communicate with others on its subnet or outside its subnet
- To communicate outside the host's own subnet, it must also know the IP address of the default router or default gateway
  - Host will forward all IP datagrams with a destination outside its own subnet to the default router
- Minimum host configuration includes
  - IP address and netmask
  - Default router address
  - DNS server addresses (will be discussed later)
- Static configuration: Administrator sets the above in the network configuration part of the host's operating system
- Dynamic configuration: Hosts use Dynamic Host Configuration Protocol (DHCP) to dynamically get the above from a server



Default router address for all hosts on subnet A is 223.1.1.4

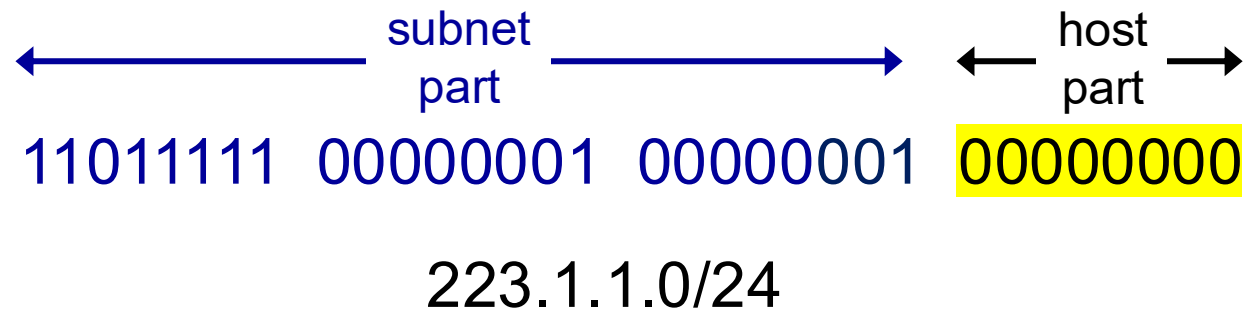
Exercise:
What should be the default router addresses for hosts on the other two subnets?

Host configuration: Exercise

- Connect to the SLIIT-STD WiFi network from your notebook or mobile
- Find the host configuration of your device
  - On iPhone devices, go to Settings -> Wi-Fi and then tap on ⓘ icon next to SLIIT-STD
  - On Android devices, go to Settings -> Wi-Fi and Network Connections and then tap on the SLIIT-STD network name
  - On notebooks go to Control Panel / System settings and see properties of the wireless network interface
- Enter your answers in Host configuration quiz on courseweb

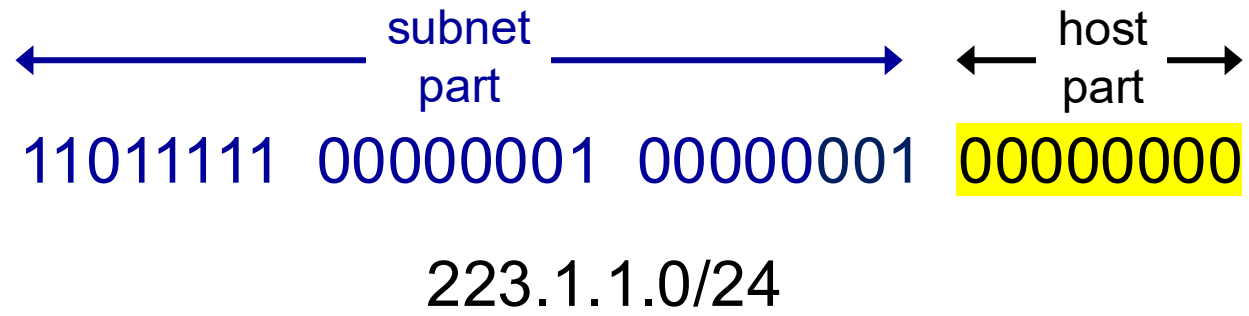
IP addresses in a subnet

- All the IP addresses used in a subnet form a contiguous range
  - Like 223.1.1.1, 223.1.1.2, 223.1.1.3, 223.1.1.4 in previous slide
- They have a common part and a varying part
- The common part is called the subnet part or the prefix, and the varying part is called the host part
- Length (number of bits) of the subnet part is called prefix length
  - Lenth of host part = 32 – prefix length
- The address range in the subnet is expressed in the notation *a.b.c.d/x* where *x* is the prefix length



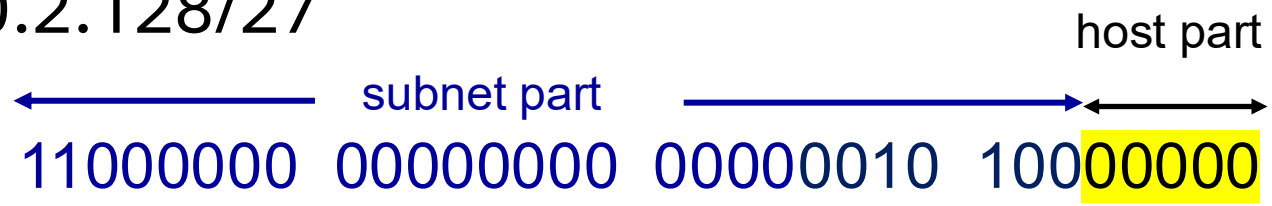
IP addresses in a subnet (ctd)

- Host part of an IP address can take any value, except
  - all 0s: reserved to specify the subnet address
  - all 1s: reserved for the broadcast address (also called the directed broadcast address) in the subnet
- The maximum number of hosts or routers in the network is therefore 2^y -2 where y is the length of the host part
- In our example
  - lowest usable value of the host part is 1 (00000001) : 223.1.1.1
  - highest usable value is 254 (11111110) : 223.1.1.254
  - broadcast address where host part is all 1s (11111111) : 223.1.1.255



IP addresses in a subnet: example 2 (method 1)

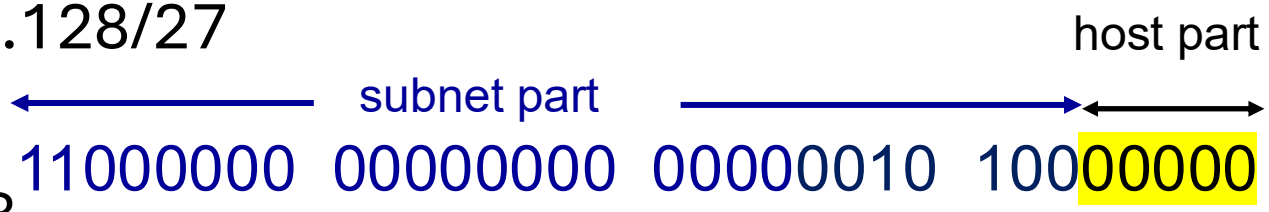
- Consider the network 192.0.2.128/27



- Prefix length = 27 bits, Host part length = 5 bits
- Network address: host part is all zeros => 192.0.2.128
  11000000 00000000 00000010 10000000
- Broadcast address: host part is all ones => 192.0.2.159
  11000000 00000000 00000010 10011111
- Lowest usable host IP address: host part = 00001 => 192.0.2.129
  11000000 00000000 00000010 10000001
- Highest usable host IP address: host part = 11110 => 192.0.2.158
  11000000 00000000 00000010 10011110

IP addresses in a subnet: example 2 (method 2)

- Consider the network 192.0.2.128/27



- Network address: 192.0.2.128
- Prefix length = 27 bits => Host part length = 5 bits
- Total number of IP addresses in range = 2^5 = 32
  - First address: Network address = 192.0.2.128
  - Last address: Network address + 32 -1 => 192.0.2.159
- Broadcast address: last address in the range => 192.0.2.159
- Number of IP addresses usable for hosts = 32 - 2 = 30
- Lowest usable host IP address
  - Address just after the network address => 192.0.2.129
- Highest usable host IP address
  - Address just below the broadcast address => 192.0.2.158

https://focnet.sliit.lk/ie2004/ipv4calc.php

Netmasks

- A netmask is an alternative way of specifying the size of a subnet
- Serves the same purpose as the prefix length
- If the prefix length is *x*, express *x* 1s followed by *32 –x* 0s in dotted decimal notation
- Examples
  - /24 means a netmask of 11111111 11111111 11111111 00000000 which is 255.255.255.0 in dotted decimal notation
  - /27 means a netmask of 11111111 11111111 11111111 11100000 which is 255.255.255.224 in dotted decimal notation
  - /22 means a netmask of 11111111 11111111 11111100 00000000 which is 255.255.252.0 in dotted decimal notation
  - /8 means a netmask of 11111111 00000000 00000000 00000000 which is 255.0.0.0 in dotted decimal notation

Prefix length, Netmask and address space

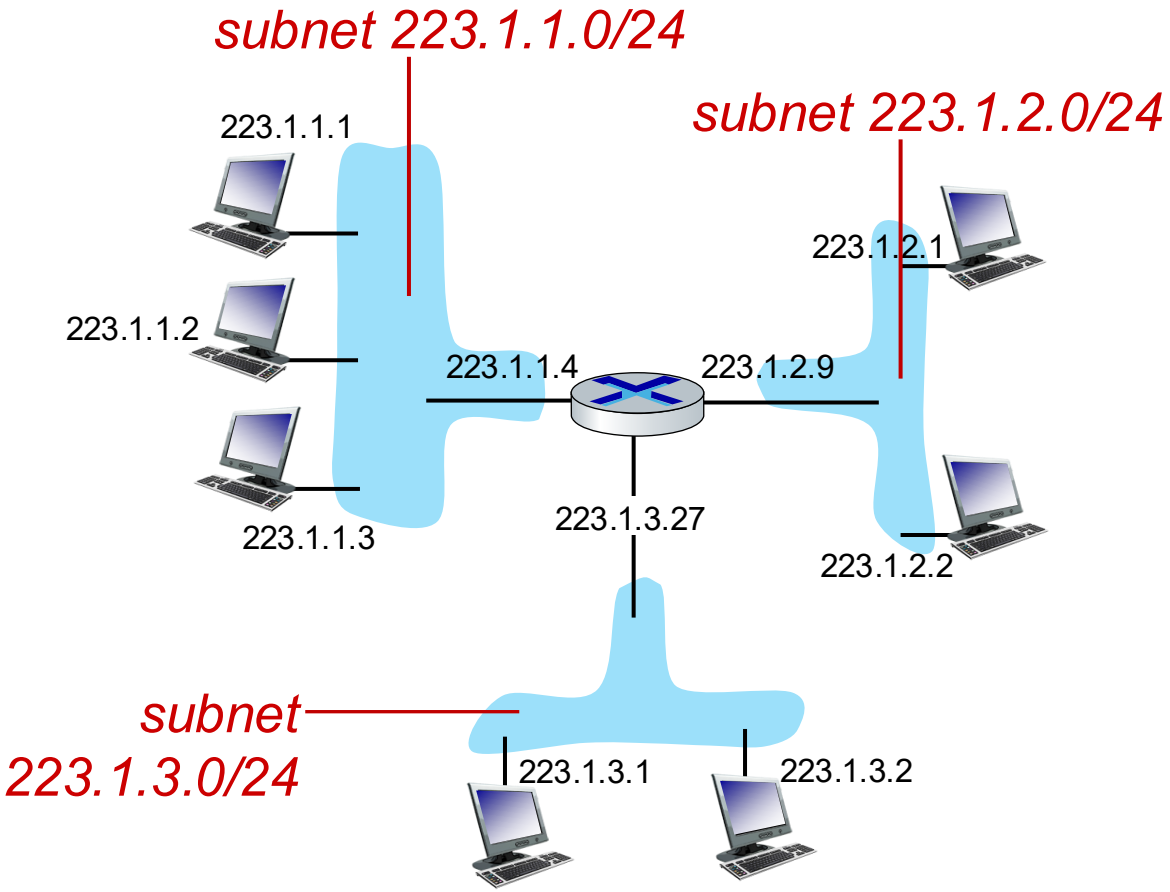
- This table shows the prefix length, netmask and the corresponding number of addresses (total and usable)

Prefix length	netmask	Total addresses	Usable addresses
/19	255.255.224.0	8192	8190
/20	255.255.240.0	4096	4094
/21	255.255.248.0	2048	2046
/22	255.255.252.0	1024	1022
/23	255.255.254.0	512	510
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2

Subnets

Recipe for defining subnets:

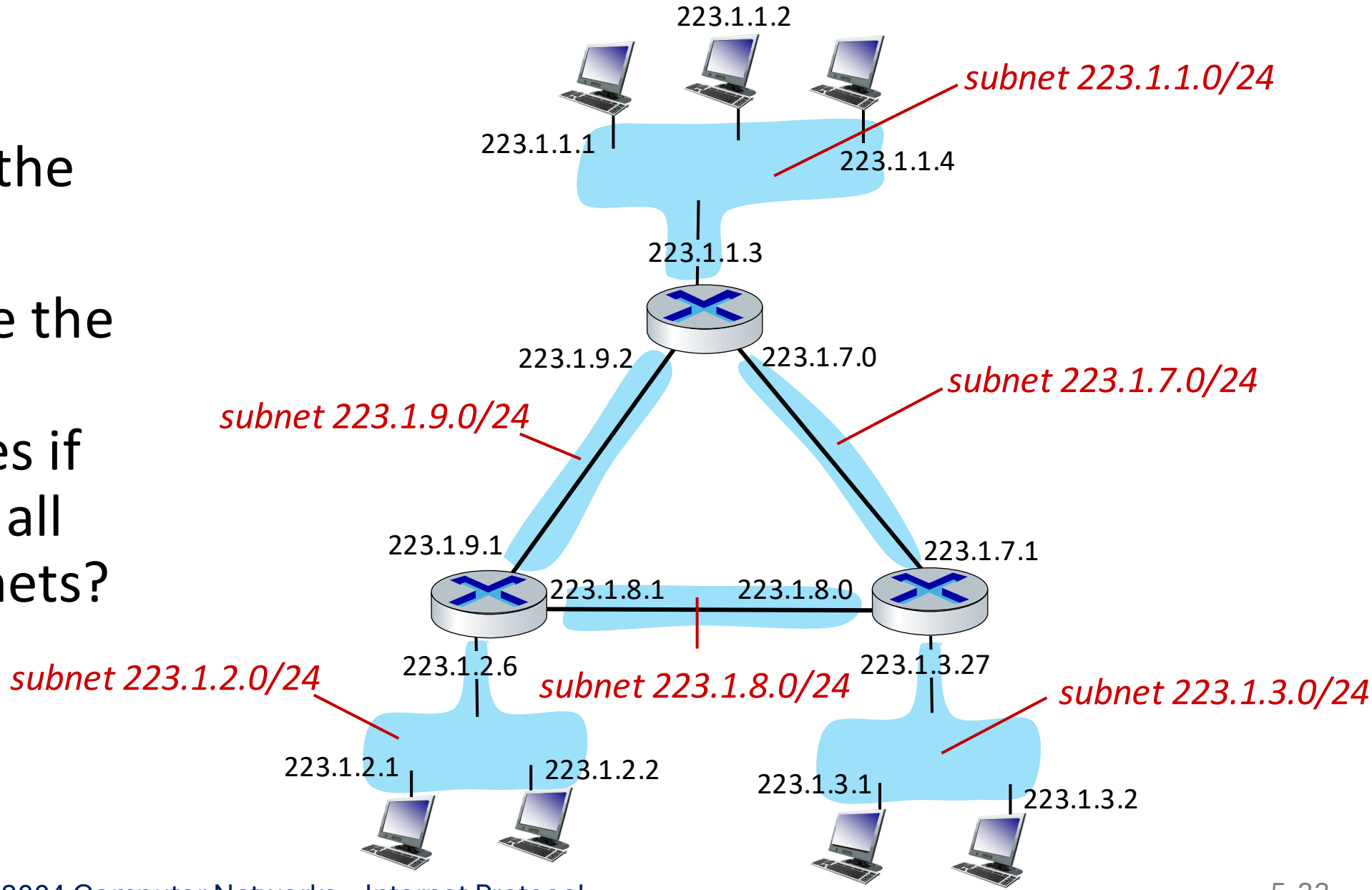
- detach each interface from its host or router, creating "islands" of isolated networks
- each isolated network is called a *subnet*



subnet mask: /24
(high-order 24 bits: subnet part of IP address)

Subnets

- Identify the subnets.
- What are the subnet addresses if they are all /24 subnets?



IP address notation

- We have seen IP address ranges expressed in the format a.b.c.d/x
- It is called the CIDR (Classless Inter Domain Routing) notation
- Any trailing zeros in the prefix (i.e. a.b.c.d) can be omitted
  - 223.1.1.0/24 may be written as 223.1.1/24
  - 10.0.0.0/8 may be written as 10/8
  - 10.0.2.0/24 can be written as 10.0.2/24 but not as 10.2/24
- The notation is used for subnet addresses and host addresses
  - If the host part is non-zero or the prefix length is 32, it is a host address
  - Otherwise, it is a subnet address
  - 223.1.1.0/24 is a subnet address as the host part is all zeros
  - 223.1.1.12/24 is a host address because the host part is non-zero
  - 0.0.0.0/32 is a host address because prefix length is 32

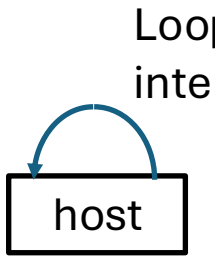
IP address exercises

- Determine the following for each of the networks given below
  - Network address
  - Netmask
  - Total number of addresses in the range
  - Number of addresses usable for hosts
  - Lowest usable host address
  - Highest usable host address
  - Broadcast address
- 10.20.40.0/22
- 10.20.50.64/26
- 10.20.60.240/30

https://focnet.sliit.lk/ie2004/ipv4calc.php

Reserved IP addresses / ranges

- 0.0.0.0/32 (all zeros): This host (me!)
  - This can be used when a host is booting up until it gets a usable IP address via DHCP or some other method
- 0.0.0.0/8: Local network (the network the host is in)
- 127.0.0.0/8: Used for loopback interfaces on a host
  - A loopback interface connects a host to itself

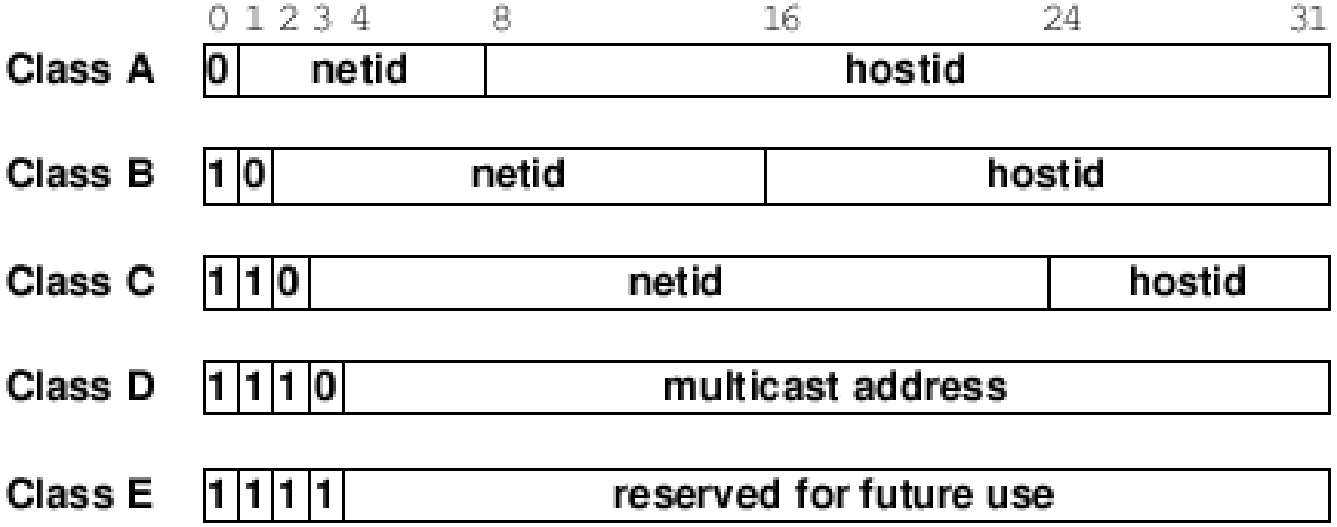


- 169.254.0.0/16: Used for link-local addresses between two hosts on a single link when no IP address is otherwise specified
- 10.0.0.0/8, 172.16.0.0/12 and192.168.0.0/16: Private IP addresses Used for local communications within a private network
- 224.0.0.0/4: Used for multicast
- 255.255.255.255/32 (all 1s): broadcast address on “this” network This is called the limited broadcast address
- Plus a few other ranges

IP address classes (obsolete)

- Historically, IP addresses were categorized into classes
- The first 1-4 bits of a network address space defined its class and implied the prefix length for the network
  - Class A: /8, Class B: /16, Class C: /24
- At present, the implied prefix length is not used
  - prefix length or netmask should be always specified using CIDR notation

Do NOT use the implied prefix length based on the address class



Broadcast addresses

- The IP protocol supports broadcasting
  - Allows a datagram to be sent to ALL hosts / routers on the subnet
- Broadcasting can be done by setting the destination address of an IP datagram to the broadcast address of the network
  - Host part of the broadcast address is all 1s
- A broadcast IP datagram must be delivered to ALL hosts / routers on the network
  - On Ethernet, the IP datagram will be encapsulated into an Ethernet frame having destination address of all 1s: ff:ff:ff:ff:ff:ff which is the Ethernet broadcast address
- The broadcast address of a subnet can be used only as the destination address in an IP datagram, not as the source address

Directed broadcast vs Limited broadcast

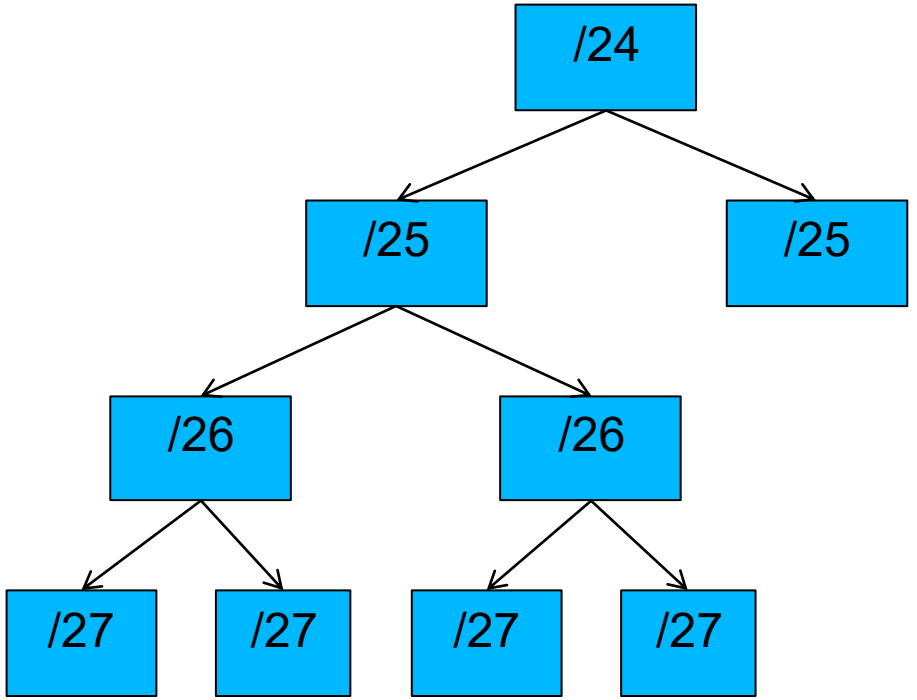
- Directed broadcast address of a network is obtained by setting the host part to all 1s
  - For the network 223.1.1.0/24 it is 223.1.1.255
  - An IP datagram originating anywhere on the Internet with a destination address of 223.1.1.255 will be broadcast on the network 223.1.1.0/24
  - The term directed means that any host on the Internet can use this address to send a broadcast datagram to this network
- Limited broadcast address is all 1s in the entire IP address (i.e. 255.255.255.255)
  - An IP datagram with limited broadcast address as the destination address is limited to that subnet
  - It does not go outside the subnet in which it originated
  - Hence the term limited

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- IPv4 addresses
- IPv4 subnetting and address allocation
- Network Address Translation (NAT)
- Internet Control Message Protocol (ICMP)
- Address resolution Protocol (ARP)

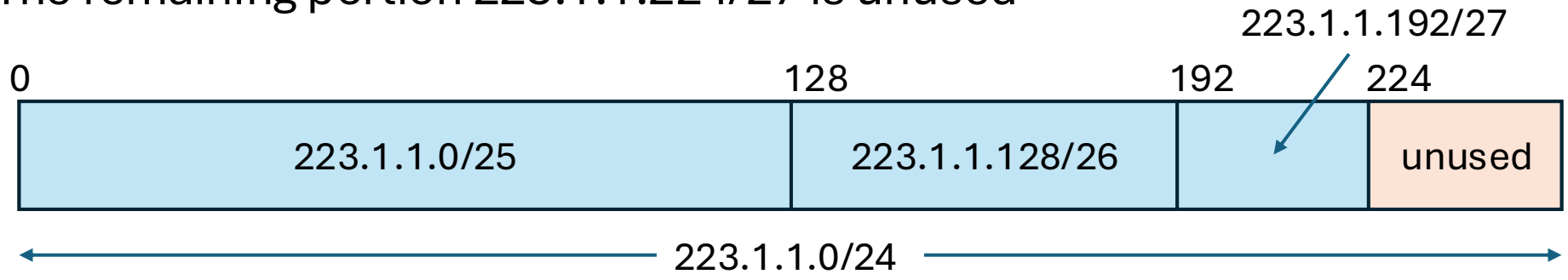
Subnetting

- Subnetting is the process of breaking up a larger IP address space into two or more smaller IP address spaces
- We can use the aggregation tree shown here to visualize the process of subnetting
  - e.g. /26 divided in half gives two /27s
  - e.g. two /25s can be combined to give a /24



Subnetting example

- We have the IP address space 223.1.1.0/24 (256 addresses), and need to subnet it to provide 3 subnets having 100, 50 and 25 hosts
  - Starting with largest subnet, find the longest prefix length that will provide the required number of hosts for the subnet (refer to table in slide 5-34)
  - For the 100 host subnet: /25 will provide 126 usable addresses
    So we use the first 128 addresses (223.1.1.0 to 223.1.1.127) for that subnet
    => 223.1.1.0/25
  - For the 50 host subnet: /26 will provide 62 usable addresses
    So we use next 64 addresses (223.1.1.128 to 223.1.1.191) for that subnet
    => 223.1.1.128/26
  - For the 25 host subnet: /27 will provide 30 usable addresses
    So we use the next 32 addresses (223.1.1.192 to 223.1.1.223) for that subnet
    => 223.1.1.192/27
  - The remaining portion 223.1.1.224/27 is unused



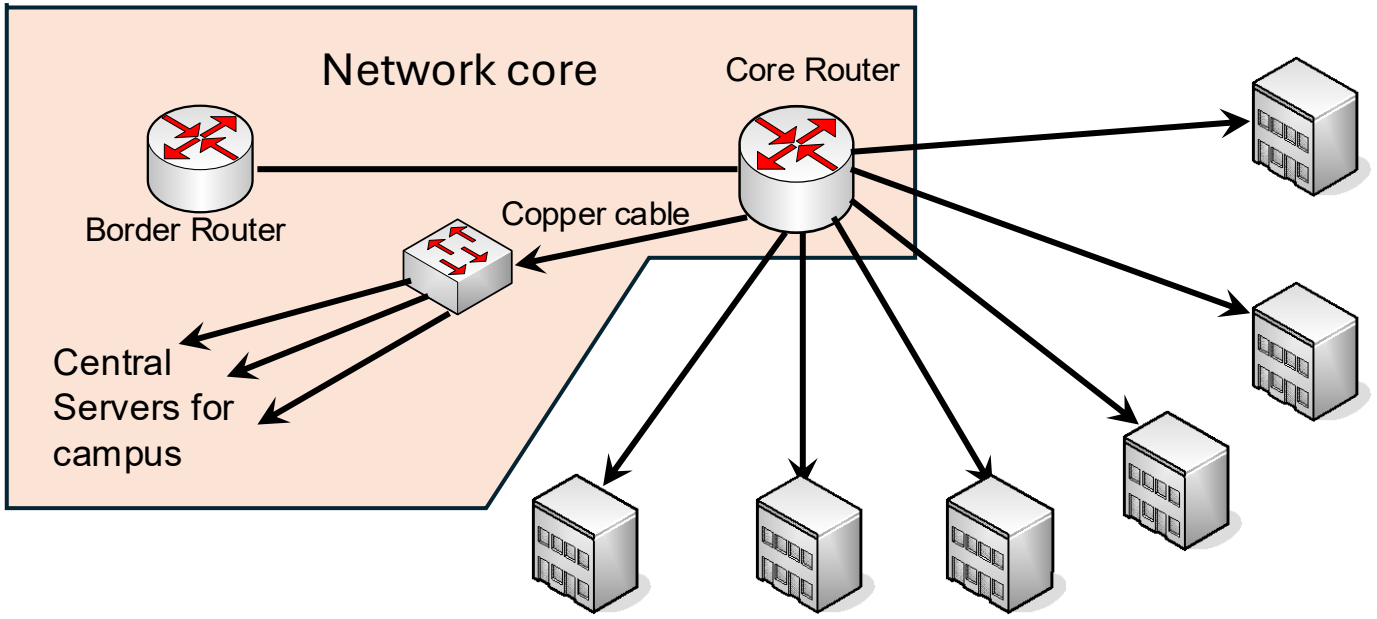
Subnetting example: Using private addresses

- If we are going to use one of the Private IP address ranges, then we have a larger address range available to us
  - 10.0.0.0/8
  - 172.16.0.0/12
  - 192.168.0.0/16
- Can use a /24 network for all networks having 254 hosts or less
- For 3 subnets having 100, 50 and 25 hosts, we can use the following address allocation in the 172.16.0.0/12 range
  - 172.16.1.0/24 for the subnet of 100 hosts
  - 172.16.2.0/24 for the subnet of 50 hosts
  - 172.16.3.0/24 for the subnet of 25 hosts

Subnetting / IP address allocation exercise

- A campus network has several buildings and a campus wide wireless network. The IP address space 223.4.8.0/21 is available for use in the campus. Reserve the first /24 of this address space for the network core and allocate IP address ranges for the subnets indicated below.

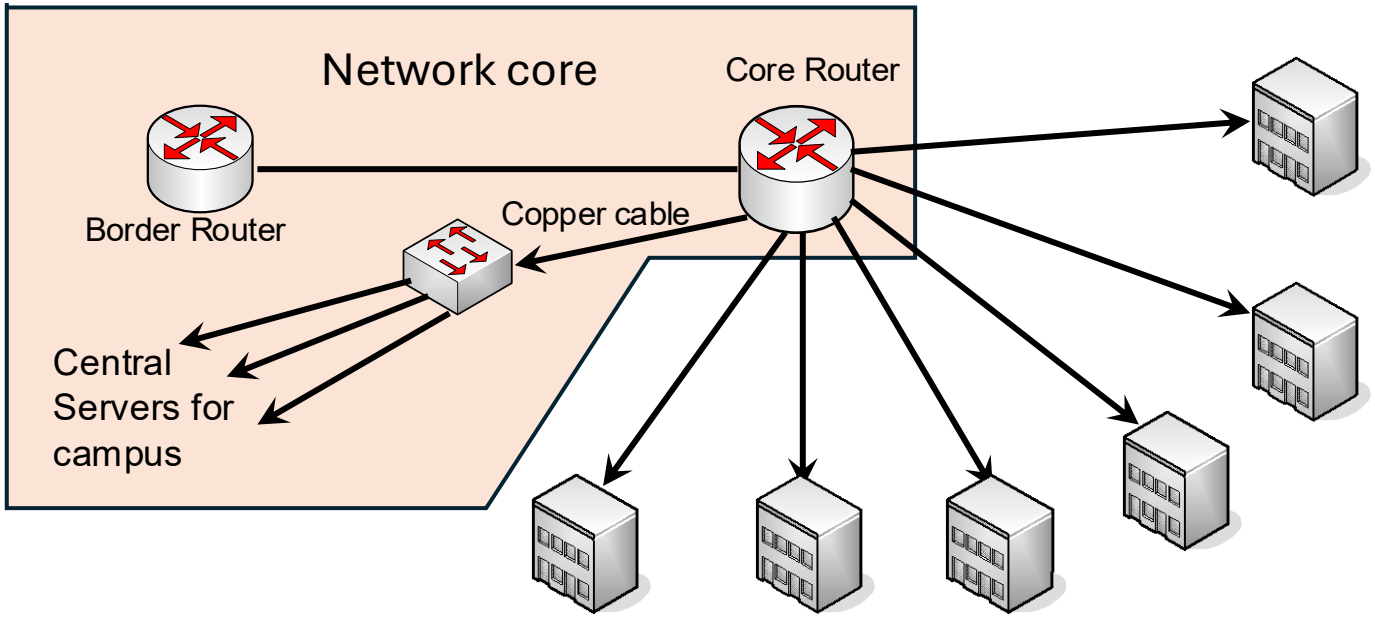
Network	Hosts
Science Building	120
Arts Building	50
Engineering Building	200
Library	80
Administration Building	40
Staff & Student Hostel	60
Wireless Network	350



Subnetting / IP address allocation exercise 2

- A campus network has several buildings and a campus wide wireless network. The private IP address space 172.16.0.0/12 is available for use in the campus. Reserve the first /24 of this address space for the network core and allocate IP address ranges for the subnets indicated below.

Network	Hosts
Science Building	120
Arts Building	50
Engineering Building	200
Library	80
Administration Building	40
Staff & Student Hostel	60
Wireless Network	350



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- Internet Control Message Protocol (ICMP)
- Address resolution Protocol (ARP)

Getting IP address space for your network

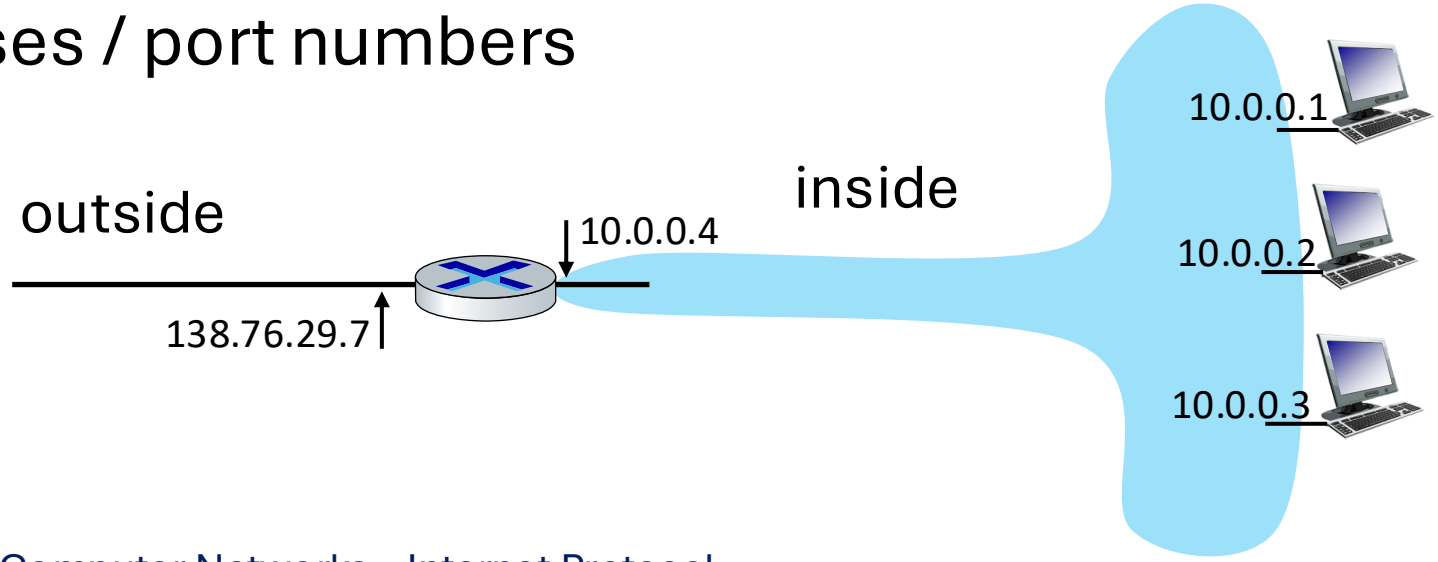
- What IP address space can you use for a particular network?
- For communications within a private network (network of an organization or a home network) you can use private address space
  - 10/8, 172.16/12 and 192.168/16
- But these private IP addresses are not usable on public Internet
- So, to communicate outside the private network, a host needs an address other than a private IP address
- Such addresses are called public IP addresses or globally routable unicast addresses

Getting IP address space for your network (2)

- You can only use a public address space allocated or assigned to you
  - By an Internet registrar (usually for an annual payment)
    - But currently the IPv4 address space is fully exhausted
  - By your Internet Service Provider (ISP)
- ISPs will assign an IP address or a set of IP addresses to a customer
  - These addresses are from an address space allocated to the ISP by an Internet registrar
  - But the ISP will usually not provide one public address for each host in your network
  - The common practice today is to use Private IP addresses within your network and use Network Address Translation (NAT) with the few public IP addresses provided by your ISP

NAT: network address translation

- Usually, devices in a local network are allocated addresses in a "private" IP address space (10/8, 172.16/12, 192.168/16)
- NAT is performed by the router connecting a private network (inside) to the public Internet (outside)
- Modifies the IP header and the TCP or UDP header to map private addresses / port numbers to public addresses / port numbers



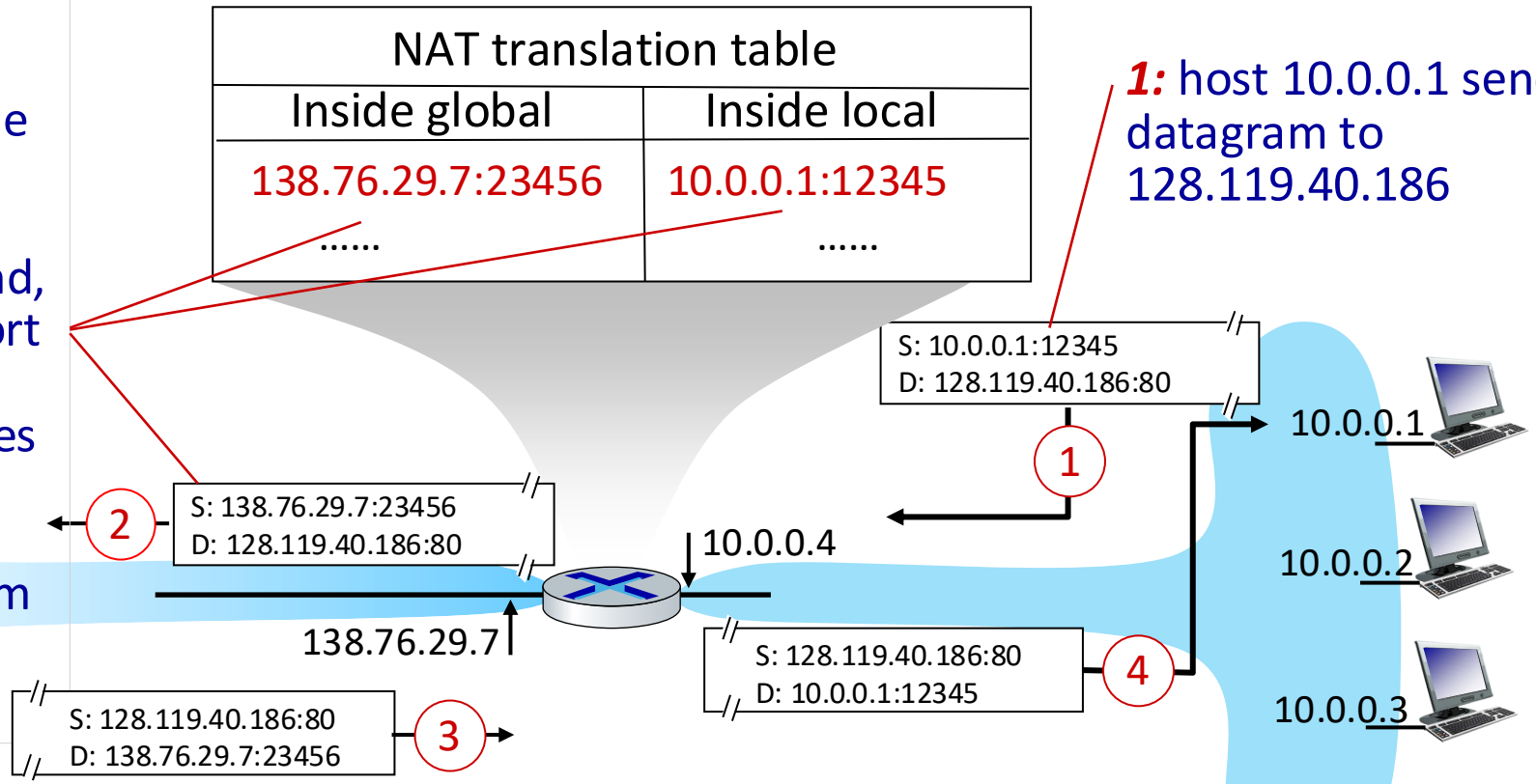
NAT: network address translation

- implementation: NAT router must (transparently):
  - outgoing datagrams: replace source IP address + port of every outgoing datagram to NAT IP address + port
    - remote clients/servers will respond using NAT IP address + port as destination address + port
  - remember (in NAT translation table) every source IP address + port to NAT IP address + port translation pair
  - incoming datagrams: replace NAT IP address + port in destination fields of every incoming datagram with corresponding source IP address + port stored in NAT table

NAT: network address translation

2: NAT router
1. Checks if a mapping already exists in the table for inside address/port combination
2. As no mapping is found, allocates an available port (e.g. 23456) for this connection*, and updates the translation table
3. Translates datagram source address/port from inside local IP / port to inside global IP / port

* NAT applied to TCP and UDP



NAT: advantages / disadvantages

- Advantages:
  - just one IP address needed from provider ISP for all devices
  - can change addresses of host in local network without notifying outside world
  - can change ISP without changing addresses of devices in local network
  - security: devices inside local net not directly addressable, visible by outside world
    - If no matching entry is found in the NAT table for an outgoing (inside → outside) packet, a port is allocated, and an entry is created
    - Incoming (outside → inside) packets are dropped if no matching entry is found
- Disadvantages
  - breaks some applications where the application layer message includes the original IP addresses and port numbers (before NAT)
  - breaks the end-to-end model of the original TCP/IP design

NAT: Exercise

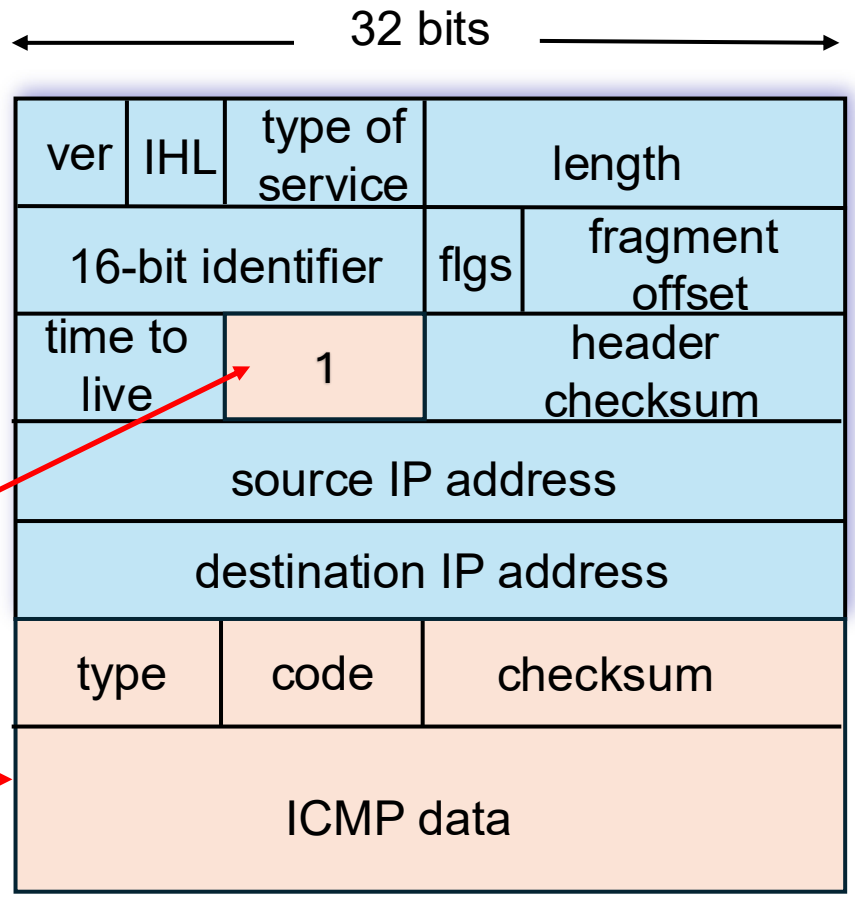
- Connect to the SLIIT-STD WiFi network from your notebook or mobile
- Find the IP address of your device
  - On iPhone devices, go to Settings -> Wi-Fi and then tap on ⓘ icon next to SLIIT-STD
  - On Android devices, go to Settings -> Wi-Fi and Network Connections and then tap on the SLIIT-STD network name
  - On notebooks go to Control Panel / System settings and see properties of the wireless network interface
- Now open a browser on your device, and go to http://ifconfig.me/
  - This web page will show you the public IP address your NAT router is using for your device (this should be same for all)
  - Note the port number (this should be unique for each)

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- Address resolution Protocol (ARP)

ICMP: internet control message protocol

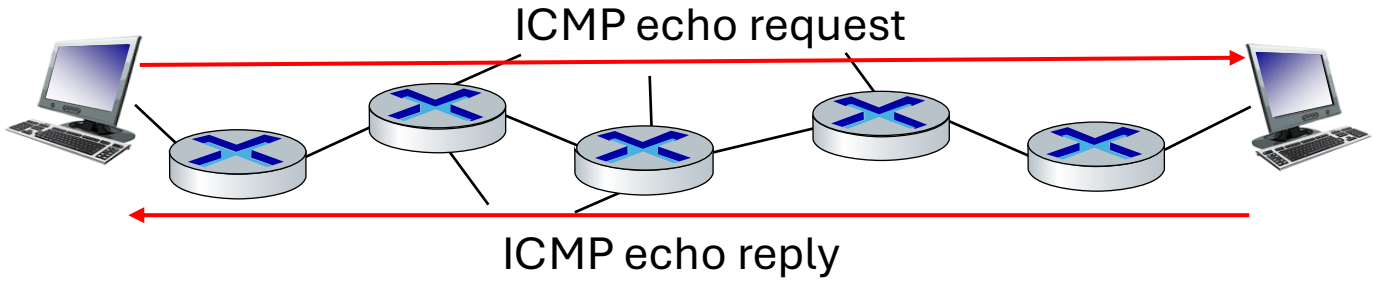
- used by hosts and routers to communicate network-level information
  - error reporting: unreachable host, network, port, protocol
  - echo request/reply (used by ping)
- network-layer "above" IP:
  - ICMP messages carried in IP datagrams with upper layer field set to 1
- *ICMP message:* type, code, checksum, and data (e.g. first 8 bytes of IP datagram causing an error)



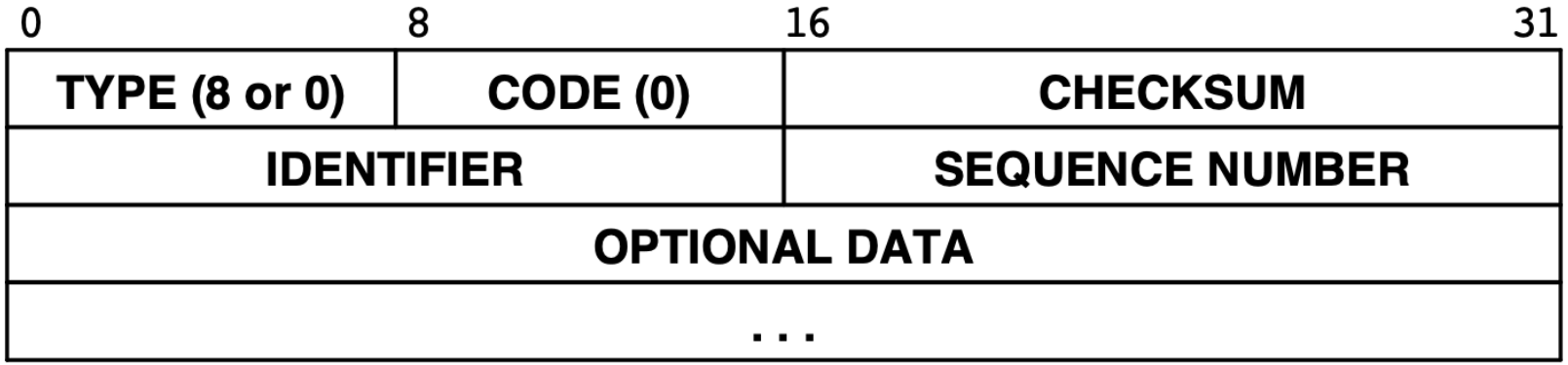
ICMP messages

Type	Code	description
0	0	echo reply (ping)
3	0	dest. network unreachable
3	1	dest host unreachable
3	2	dest protocol unreachable
3	3	dest port unreachable
3	4	fragmentation needed but DF bit is set
3	6	dest network unknown
3	7	dest host unknown
3	9	communication with destination network is administratively prohibited
3	10	communication with destination host is administratively prohibited
8	0	echo request (ping)
9	0	route advertisement
10	0	router discovery
11	0	TTL expired
12	0	bad IP header

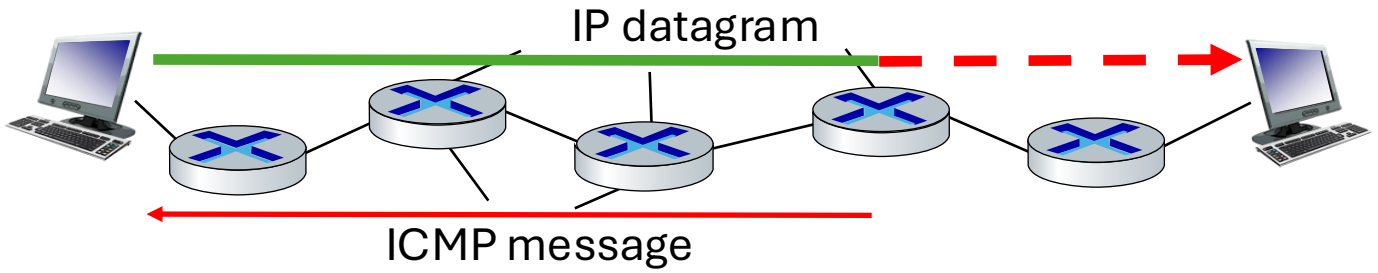
ICMP echo request / reply (Ping)



- Source sends ICMP echo request to destination (type = 8, code = 0)
- Destination replies with ICMP echo reply (type =0, code = 0)
- Identifier and sequence number used to match requests and replies



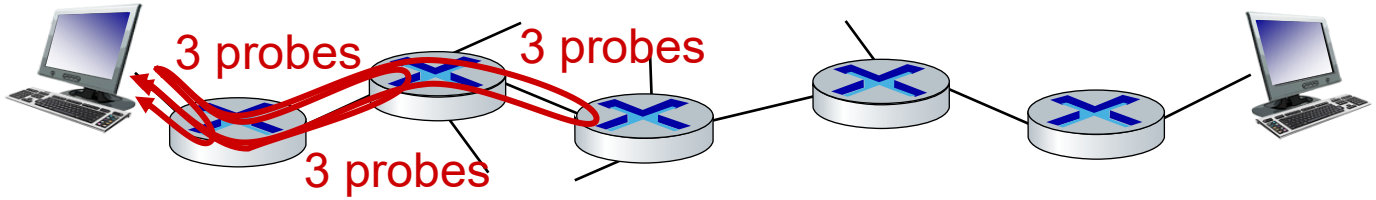
ICMP Destination unreachable



- Usually sent by an intermediate router, when an IP datagram cannot be sent to the intended destination
- Type =3, Code indicates the cause of the error
- The header and the first 8 bytes of the original datagram appended as data

0	8	16	31
TYPE (3)	CODE (0-12)	CHECKSUM	
UNUSED (MUST BE ZERO)			
INTERNET HEADER + FIRST 64 BITS OF DATAGRAM			
. . .			

Traceroute and ICMP



- source sends sets of UDP segments to destination
  - 1st set has TTL =1, 2nd set has TTL=2, etc.
- datagram in *n*th set arrives to nth router:
  - router discards datagram and sends ICMP message to source (type 11, code 0)
  - ICMP message possibly includes name of router & IP address
- when ICMP replies arrive at source: record RTTs

stopping criteria:

- UDP segment eventually arrives at destination host
- destination returns ICMP "port unreachable" message (type 3, code 3)
- source stops

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- Network Address Translation (NAT)
- Internet Control Message Protocol (ICMP)
- Dynamic Host Configuration Protocol (DHCP)
- Address Resolution Protocol (ARP)

Host configuration

- For a HOST on the Internet to communicate with any other host on the Internet, it should have at least
  - A globally unique IP address (and subnet information)
  - Default router which will route IP datagrams to/from outside host's own network
  - A recursive DNS server (RDNSS) which will help in name to address translation (DNS resolution)
- Host configuration can be static or dynamic
  - Static configuration in OS, Firmware
  - Dynamic configuration using DHCP

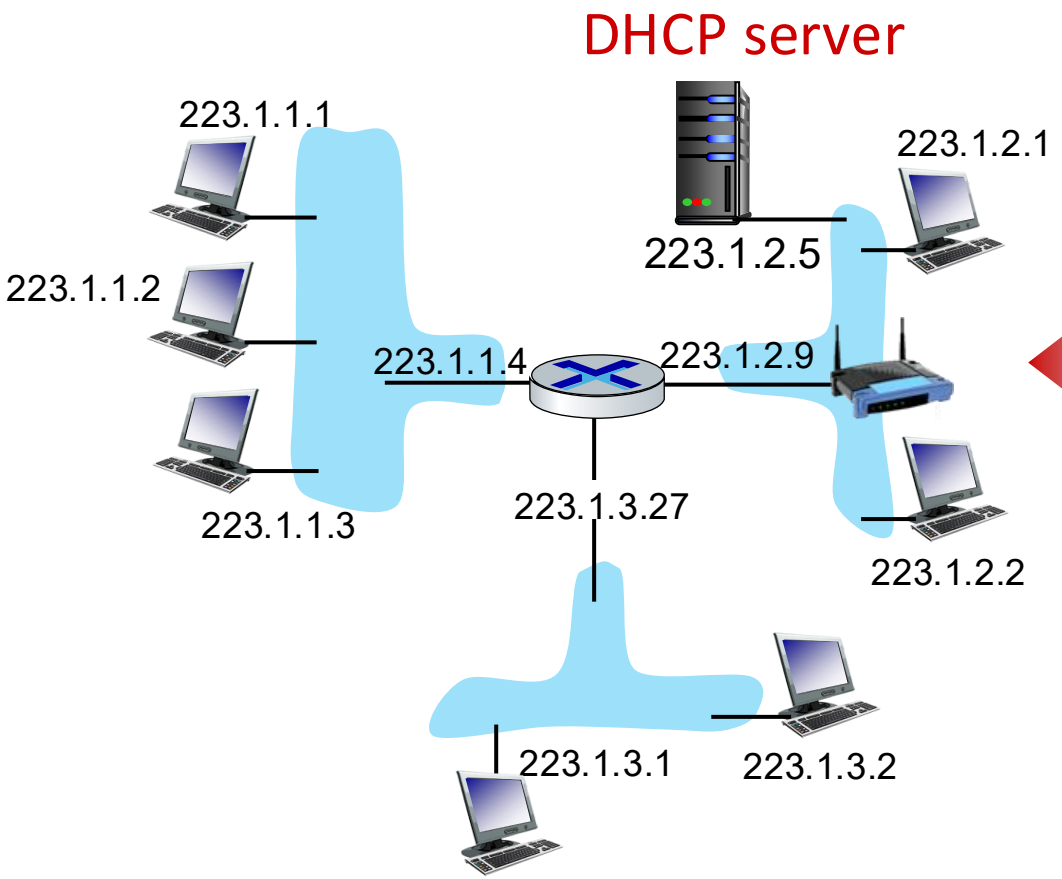
Dynamic Host Configuration Protocol (DHCP)

- Primarily used to provide clients with IP addresses when the “join” a network
- Can provide other information such as netmask, default router address, DNS server addresses, etc
  - via DHCP options
- DHCPv4 is for IPv4, DHCPv6 is for IPv6
  - DHCPv6 functionality somewhat different from DHCPv4
- We will only talk about DHCPv4 in this module
- In this slide set, DHCP means DHCPv4
- DHCP is an application layer protocol
  - Using UDP ports 67/547 (server) and 68/546 (client)

DHCP basics

- A DHCP server dynamically assigns, or leases, IPv4 addresses to client hosts from a pool of addresses
  - for a limited period of time chosen by the server (lease period)
  - or until the client no longer needs the address
- Lease period can be anywhere from a few hours to a week or more
- When the lease expires, the client must renew the lease
  - Server may re-assign the same address or assign a different address
- allows reuse of addresses
  - Client hosts only hold address while connected / on
- Specially useful for mobile users who join / leave networks often

DHCP clients and servers



Typically, DHCP server will be included in a router, serving all subnets to which router is attached

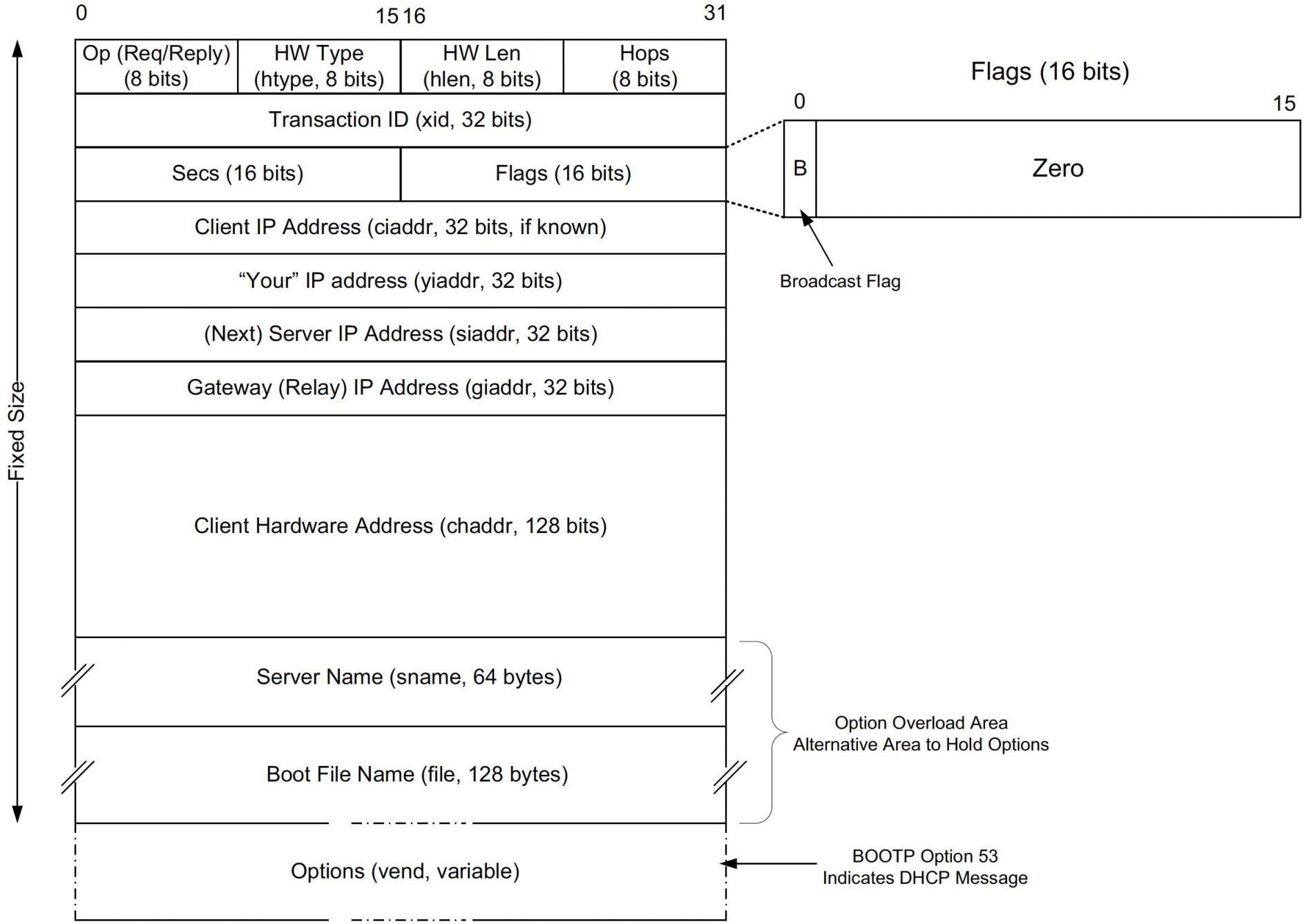


arriving DHCP client needs address in this network

DHCP messages

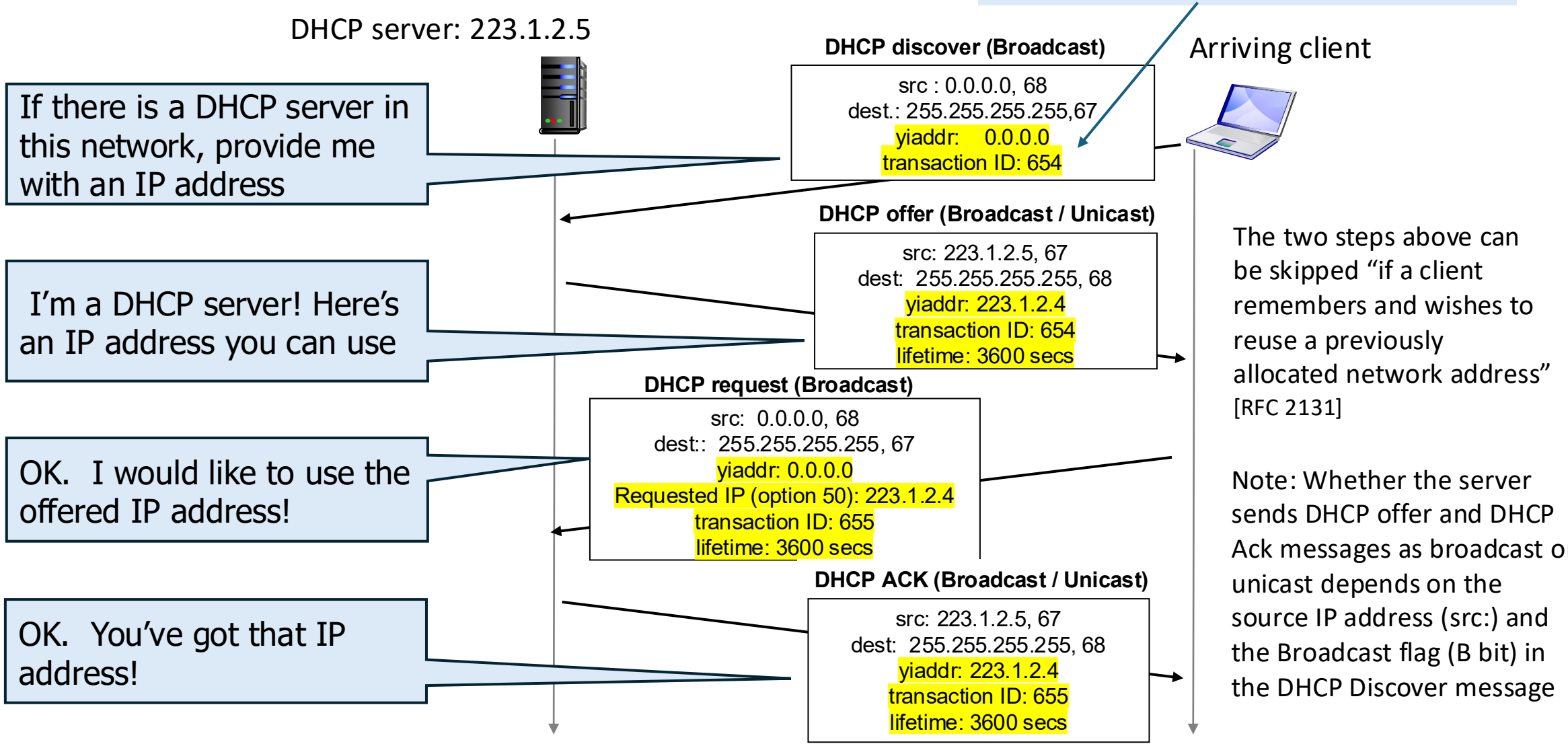
1. Client host broadcasts DHCP discover message
   - Destination address: 255.255.255.255 (limited broadcast address)
2. DHCP server(s) responds with DHCP offer message
   - Offers an IP address (multiple servers may offer IP addresses)
   - Also sent to the limited broadcast address (why?)
   - Because the host does not have its own IP address yet
3. Client host requests IP address: DHCP request message
   - Client selects an offer, confirms request of offered address from server
   - Also sent to the limited broadcast address
4. DHCP server sends address: DHCP ack message
   - Maybe sent to limited broadcast address or unicast address of client host

DHCPv4 Message Format



DHCP client-server scenario

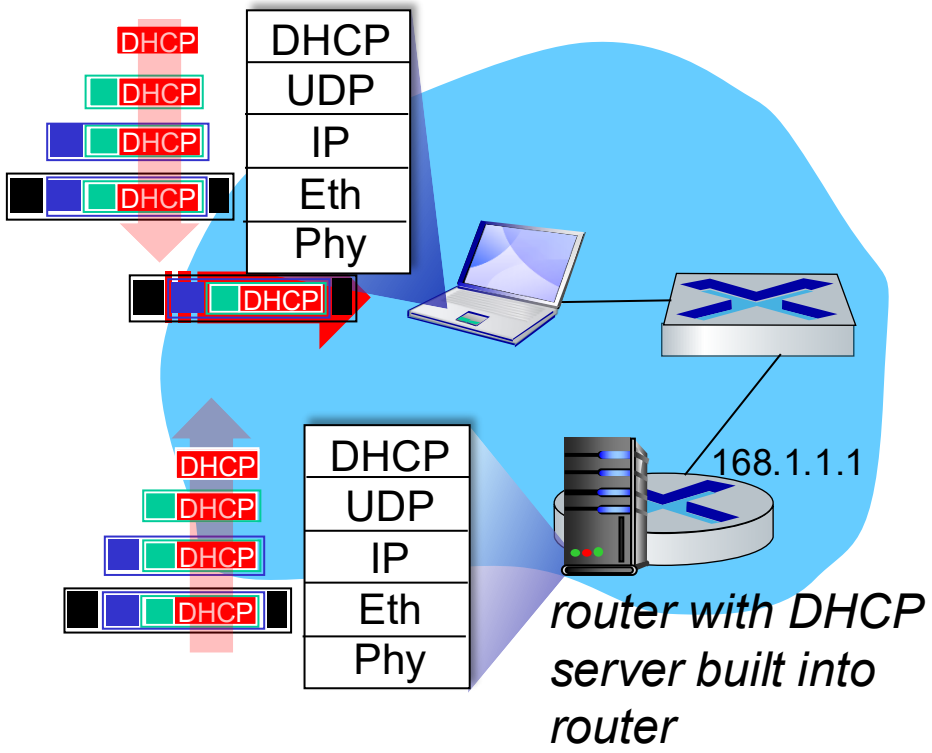
Fields of DHCP message highlighted
Rest from IP, UDP headers



The two steps above can be skipped "if a client remembers and wishes to reuse a previously allocated network address" [RFC 2131]

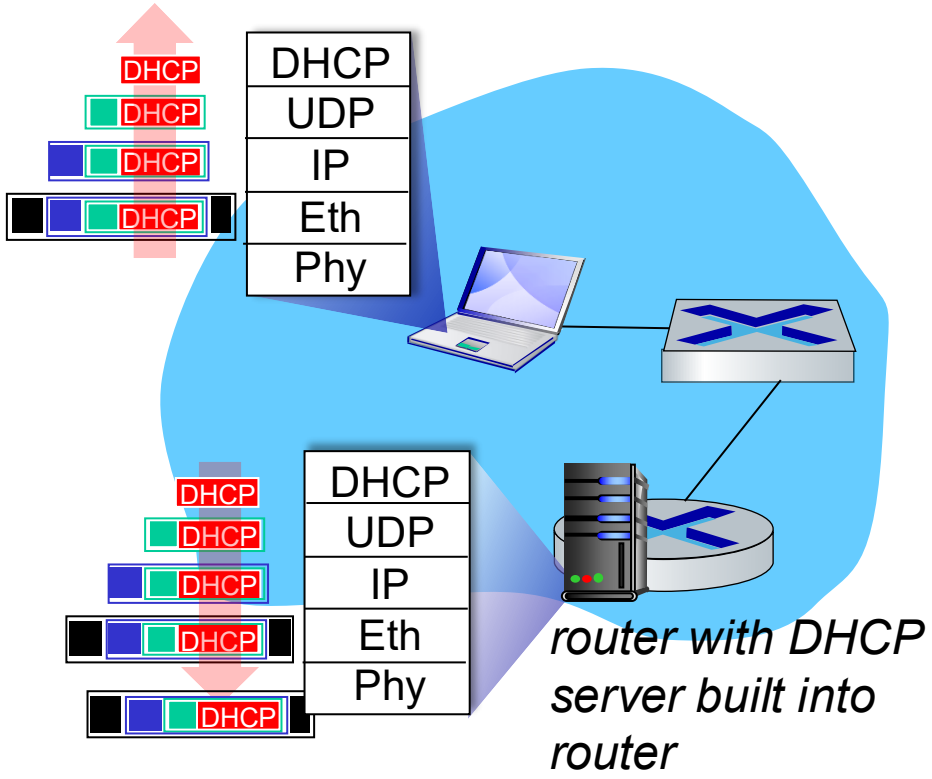
Note: Whether the server sends DHCP offer and DHCP Ack messages as broadcast or unicast depends on the source IP address (src:) and the Broadcast flag (B bit) in the DHCP Discover message

DHCP: example



- Connecting laptop will use DHCP to get IP address, address of first-hop router, address of DNS server.
- DHCP REQUEST message encapsulated in UDP, encapsulated in IP, encapsulated in Ethernet
- Ethernet frame broadcast (dest: FF:FF:FF:FF:FF:FF) on LAN, received at router running DHCP server
- Ethernet decapsulated to IP, IP decapsulated to UDP, UDP decapsulated to DHCP

DHCP: example

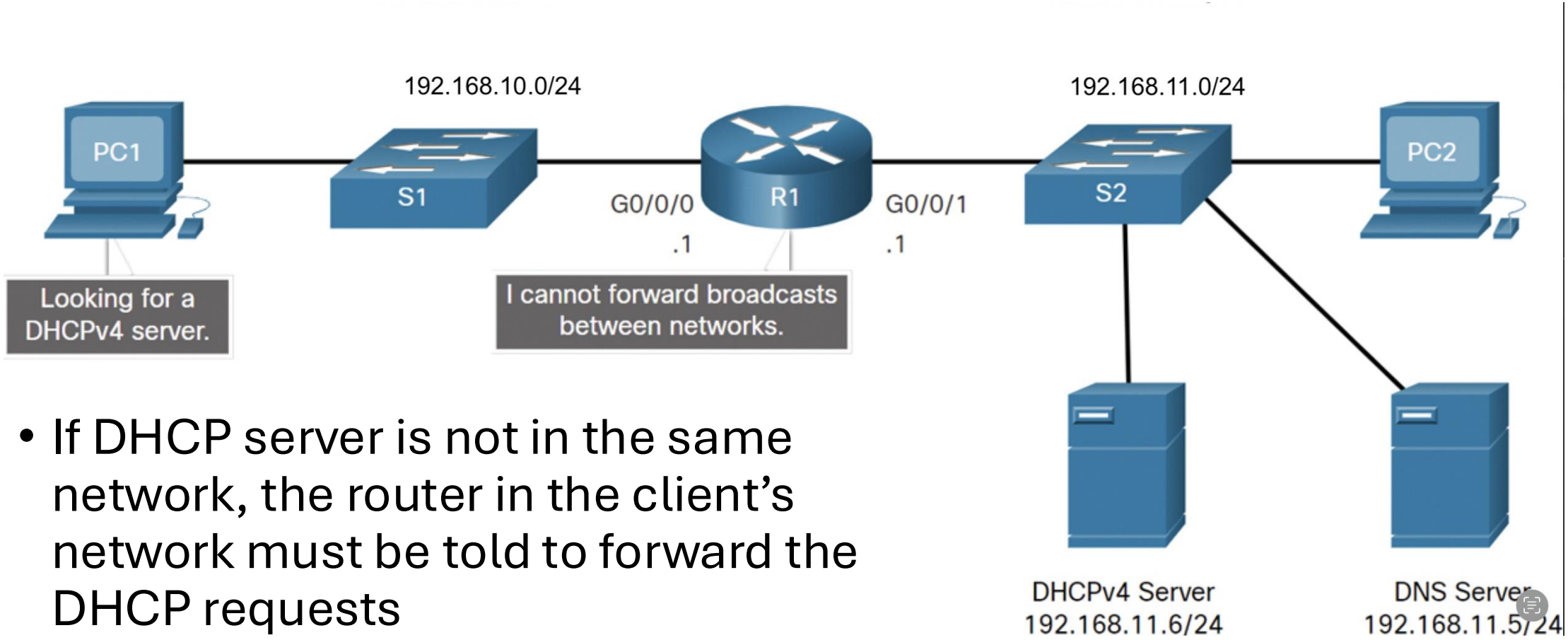


- DHCP server sends the DHCP ACK containing client's IP address, IP address of first-hop router for client, name & IP address of DNS server
- DHCP server reply encapsulated and forwarded to client, de-capsulated up to DHCP at client
- client now knows its IP address, name and IP address of DNS server, IP address of its first-hop router

DHCPv4 Relay

- In a complex hierarchical network, enterprise servers (DHCP, DNS, TFTP, FTP etc) are located in a server farm.
- Network clients are not on the same subnet as the DHCP servers.
- In order to locate the DHCP servers and receive services, clients mostly use broadcast messages
- If DHCP server is not in the same network, the router in the client's network must be told to forward the DHCP requests

DHCPv4 Relay



- If DHCP server is not in the same network, the router in the client's network must be told to forward the DHCP requests

DHCPv4 Relay

- A Cisco IOS helper address is configured so that the router acts as a relay agent forwarding the message to the DHCPv4 server.
- The interface on Router receiving the broadcast is configured with the ip helper-address interface configuration mode command.
- The address of the DHCPv4 server is configured as the only parameter.
- When Router has been configured as a DHCPv4 relay agent, it accepts broadcast requests for the DHCPv4 service and then forwards those requests as unicast to the IPv4 address of DHCP server.

DHCPv4 Relay

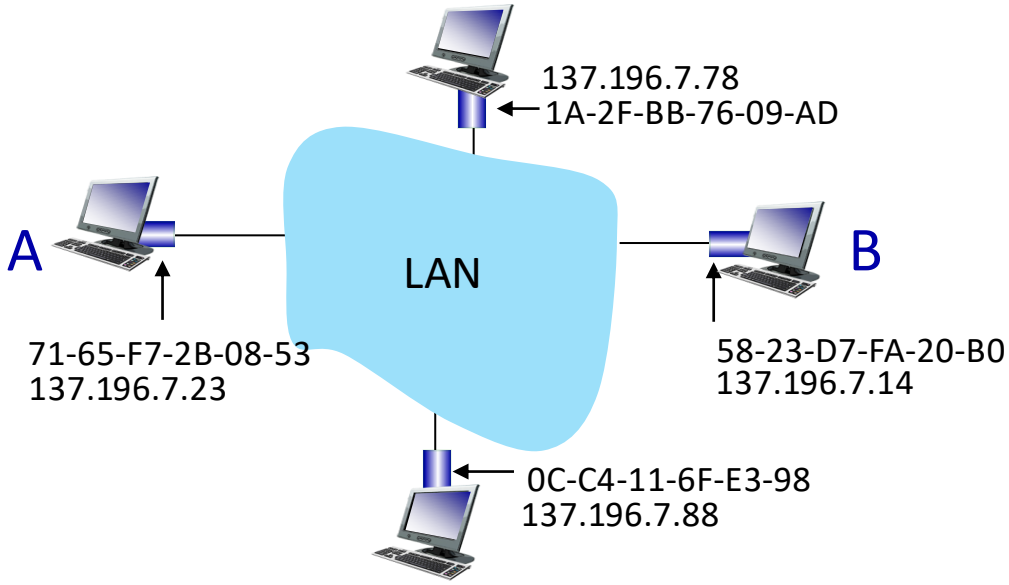
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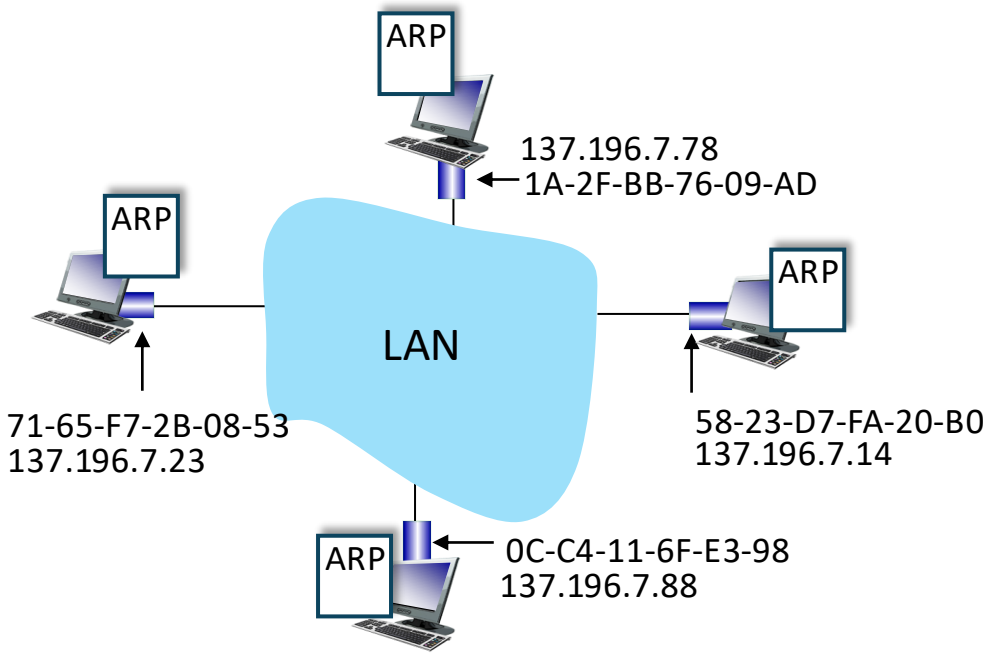
ARP: address resolution protocol

- Host A has an IP datagram to send to host B on local LAN
  - IP source: 137.196.7.3
  - IP Destination: 137.196.7.14
- Link layer of host A must encapsulate this IP datagram in an Ethernet frame
  - Source MAC: 71-65-F7-2B-08-53
  - Destination MAC: ?
- ARP protocol is used to find the MAC address of a host with a known IP address
- ARP is considered a network layer protocol because ARP messages are encapsulated inside link layer frames



ARP Table

- Each host maintains an ARP table conatining MAC addresses leanred with the ARP protocol

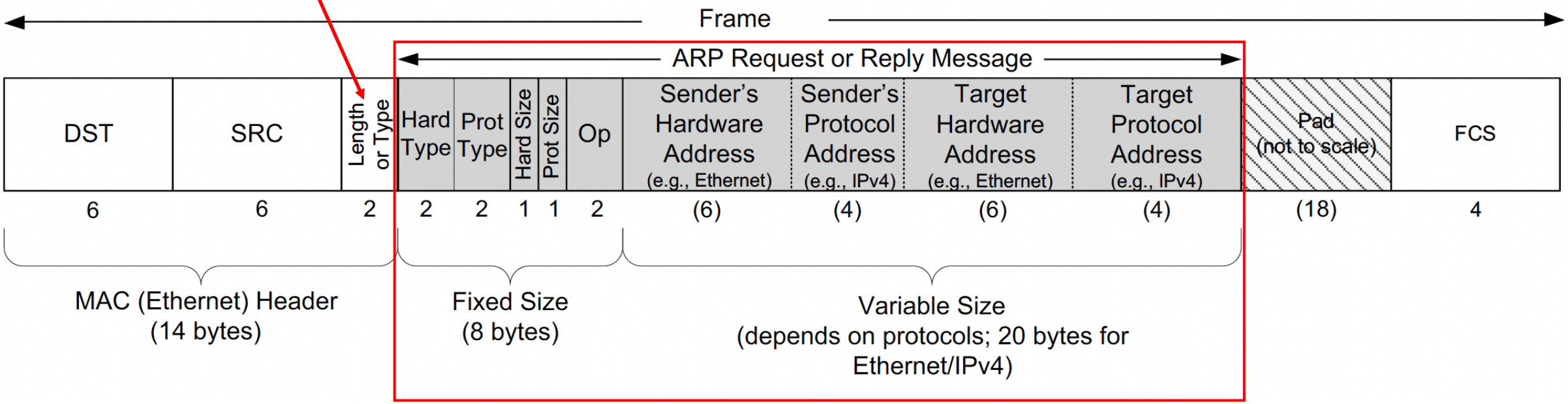


IP addr	MAC addr	TTL

- Fields in an ARP table
  - IP address
  - MAC address
  - TTL (Time To Live): time after which address mapping will be forgotten (typically 1200 seconds = 20 minutes)

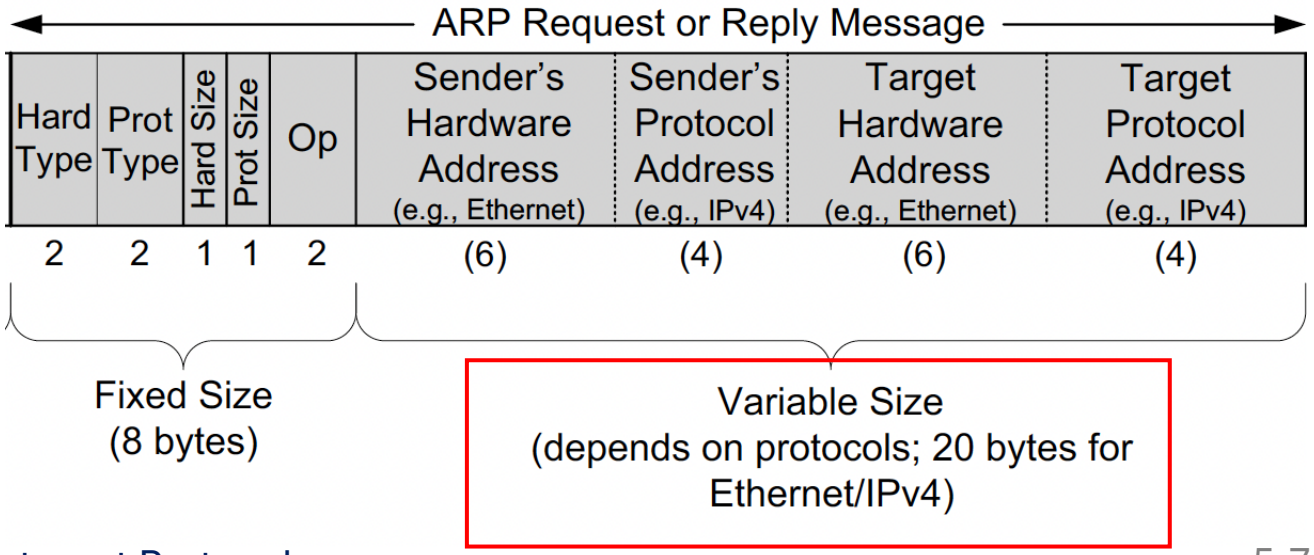
ARP Message format

- ARP messages are directly encapsulated inside Ethernet frames
  - Type = 0806 (Hexadecimal)



ARP Message format (ctd)

- Hardware type: 0001 for Ethernet
- Protocol type: 0800 for IPv4
- Hardware size (Length of hardware address): 6 for Ethernet
- Protocol size (Length of IP address): 4 for IPv4
- Opcode: 1 for request, 2 for reply, ...
- Sender HW address
- Sender IP address
- Target HW address
- Target IP address



ARP protocol in action

example: A wants to send datagram to B

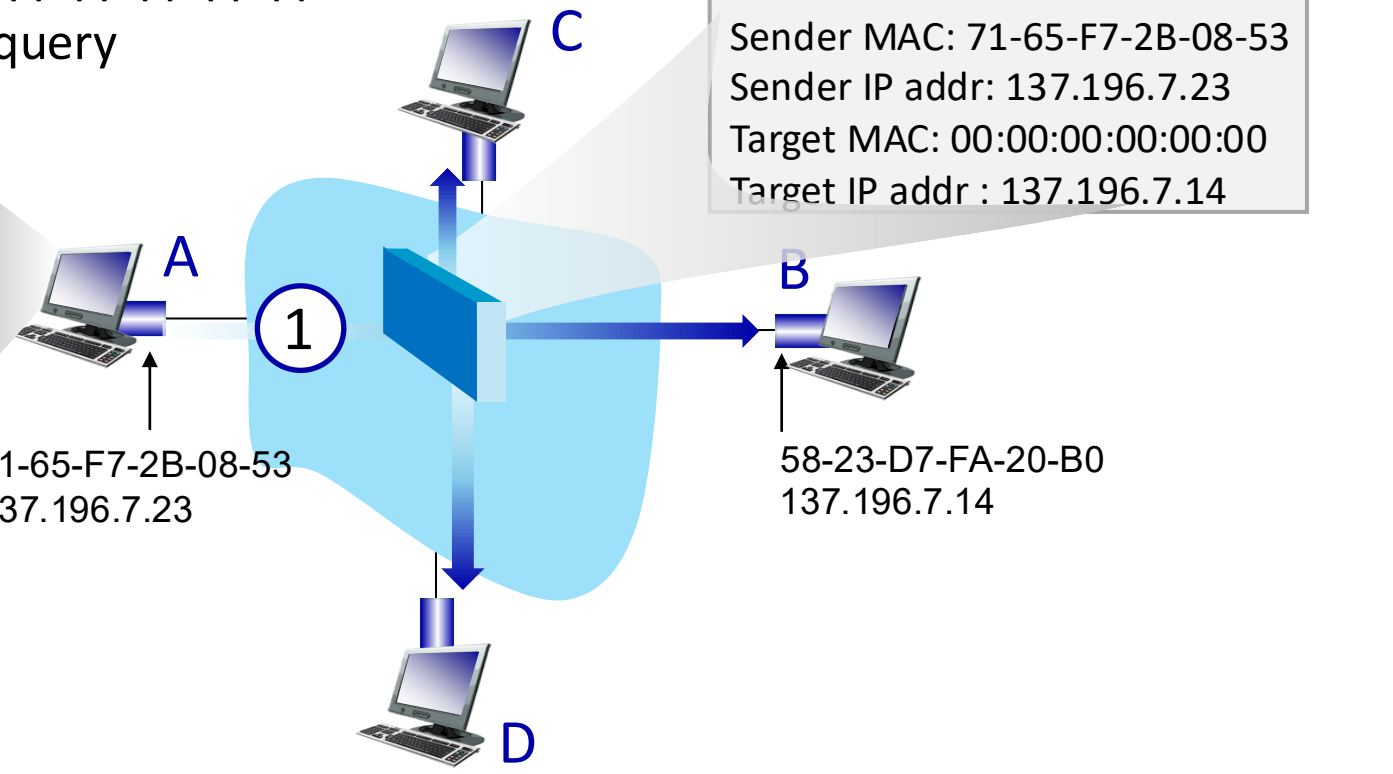
- B's MAC address not in A's ARP table, so A uses ARP to find B's MAC address

A broadcasts ARP query, containing B's IP addr

1.
   - destination MAC address = FF-FF-FF-FF-FF-FF
   - all nodes on LAN receive ARP query

ARP table in A

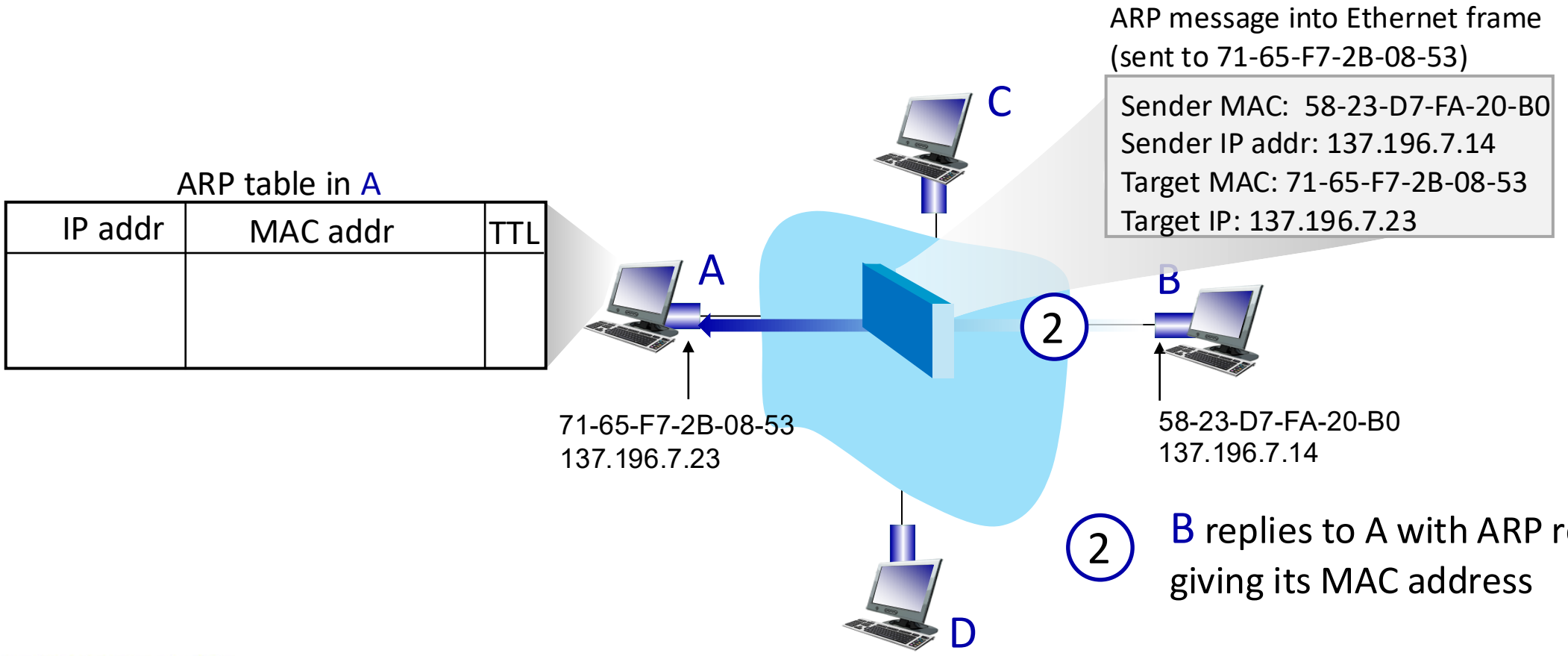
IP addr	MAC addr	TTL



ARP protocol in action

example: A wants to send datagram to B

- B's MAC address not in A's ARP table, so A uses ARP to find B's MAC address



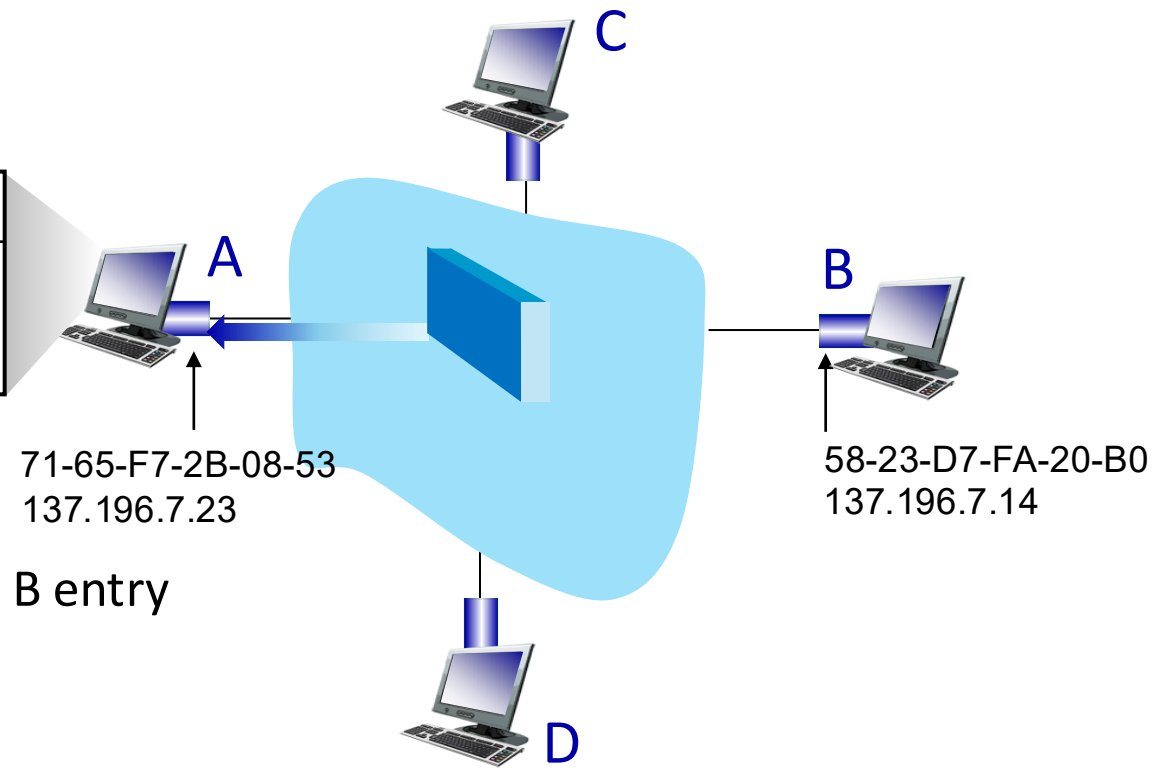
ARP protocol in action

example: A wants to send datagram to B

- B's MAC address not in A's ARP table, so A uses ARP to find B's MAC address

ARP table in A

IP addr	MAC addr	TTL
137.196.7.14	58-23-D7-FA-20-B0	500



③ A receives B's reply, adds B entry into its local ARP table

Key terms and concepts

- Routing vs forwarding
- IP protocol
- Fragmentation and reassembly
- IP addressing and subnetting
- Network Address Translation (NAT)
- Internet Control Messages
- Link layer Address Resolution

References

- Chapter 4 - The Network Layer: Data Plane
  James F. Kurose and Keith W. Ross, Computer Networking – A Top-Down Approach, (8th Edition), Pearson, 2020
- Chapter 4 - IPv6 Address Format and Address Types
  Graziani, R., IPv6 Fundamentals: A Straightforward Approach to Understanding IPv6 (2nd Edition), Cisco Press, 2017

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